

FASHION TRENDZ E-COMMERCE PLATFORM

DATABASE DESIGN DOCUMENT

Field	Value
Project Name	Fashion Trendz E-Commerce Platform
Document	Fashion Trendz Database Schema
Version	1.0
Date	05 August 2026
Database Engine	PostgreSQL 18
Total Tables	105 (41 implemented · 64 proposed)
Framework Tables	8 (Django & SimpleJWT internals — documented in Appendix A)
Total Stored Procedures	20
Total Indexes	47 existing + 18 recommended
Total Foreign Keys	63 implemented + 71 proposed
Source of Truth	Django model metadata, introspected from the live migration state
Prepared For	Fashion Trendz Database Engineering Team
Prepared By	Lead Database Architect
Classification	Confidential — Internal Engineering

This document describes the physical database design for the Fashion Trendz platform. The implemented tables are derived directly from the shipped Django backend by introspecting model metadata against the applied migration state — not transcribed by hand — so the column definitions, constraints and indexes below are the ones actually present in the database. Proposed tables are clearly marked and represent new build required to satisfy the full product brief.

OVERVIEW — COMPLETE TABLE INVENTORY

Every table in the Fashion Trendz database, numbered and grouped by module. Status distinguishes what exists today from what this document specifies for build.

#	Table Name	Description	Module	Section	Status
1	users_user	Customer and staff accounts	Authentication & User Management	Users & Auth	IMPLEMENTED
2	users_address	Delivery and billing addresses belonging to a customer	Authentication & User Management	Users & Auth	IMPLEMENTED
3	users_user_groups	Bridge assigning users to Django auth groups	Authentication & User Management	Users & Auth	IMPLEMENTED
4	users_user_user_permissions	Bridge granting individual Django permissions directly to a user	Authentication & User Management	Users & Auth	IMPLEMENTED
5	catalog_category	Top level of the merchandising tree (Women, Men, Kids, Accessories...)	Catalog Taxonomy	Catalog	IMPLEMENTED
6	catalog_subcategory	Second level of the tree (Dresses, Shirts, Sarees...)	Catalog Taxonomy	Catalog	IMPLEMENTED
7	catalog_brand	Brand master	Catalog Taxonomy	Catalog	IMPLEMENTED
8	catalog_brand_categories	Bridge: which categories a brand sells into	Catalog Taxonomy	Catalog	IMPLEMENTED
9	catalog_collection	Curated merchandising groupings — New Arrivals, Trending, Luxury, Editor's Picks, Best Sellers, Seasonal	Catalog Taxonomy	Catalog	IMPLEMENTED
10	catalog_collection_categories	Bridge scoping a collection to one or more categories	Catalog Taxonomy	Catalog	IMPLEMENTED
11	products_product	The master product record and the widest table in the schema at 57 columns	Product Catalog	Products	IMPLEMENTED
12	products_producttag	Free-form merchandising labels (Festive, Office Wear, Monsoon Ready)	Product Catalog	Products	IMPLEMENTED
13	products_product_tags	Bridge assigning merchandising tags to products	Product Catalog	Products	IMPLEMENTED
14	products_productimage	Product gallery	Product Catalog	Products	IMPLEMENTED
15	products_productvariant	The sellable unit: one row per colour and size	Product Catalog	Products	IMPLEMENTED
16	products_productattribute	Arbitrary key/value facts for filtering	Product Catalog	Products	IMPLEMENTED
17	products_productspecification	Ordered display-only specification rows (Fabric, Fit, Wash Care)	Product Catalog	Products	IMPLEMENTED
18	wishlist_wishlist	One wishlist per signed-in customer	Wishlist & Cart	Wishlist & Cart	IMPLEMENTED
19	wishlist_wishlistitem	One saved product	Wishlist & Cart	Wishlist & Cart	IMPLEMENTED
20	cart_cart	A shopping bag owned by either a signed-in customer or an anonymous session, never both and never neither — a CH...	Wishlist & Cart	Wishlist & Cart	IMPLEMENTED
21	cart_cartitem	One bag line	Wishlist & Cart	Wishlist & Cart	IMPLEMENTED
22	orders_order	The order header and the financial record of a sale	Orders & Fulfilment	Orders	IMPLEMENTED
23	orders_orderitem	One purchased line, fully snapshotted	Orders & Fulfilment	Orders	IMPLEMENTED
24	orders_orderstatushistory	Append-only audit of every state transition	Orders & Fulfilment	Orders	IMPLEMENTED
25	orders_shipment	One dispatched parcel with courier and tracking reference	Orders & Fulfilment	Orders	IMPLEMENTED
26	coupons_coupon	Discount code definition: flat, percentage or free shipping, with a minimum cart value, an optional cap, global ...	Promotions & Coupons	Coupons	IMPLEMENTED
27	coupons_coupon_categories	Bridge restricting a coupon to specific categories	Promotions & Coupons	Coupons	IMPLEMENTED
28	coupons_coupon_brands	Bridge restricting a coupon to specific brands	Promotions & Coupons	Coupons	IMPLEMENTED
29	coupons_coupon_products	Bridge restricting a coupon to specific products	Promotions & Coupons	Coupons	IMPLEMENTED
30	coupons_couponusage	Redemption ledger — the enforcement point for per-customer limits	Promotions & Coupons	Coupons	IMPLEMENTED
31	payments_payment	One payment intent against an order	Payments & Refunds	Payments	IMPLEMENTED
32	payments_paymentattempt	One try within a payment intent	Payments & Refunds	Payments	IMPLEMENTED
33	payments_refund	Money returned, whole or partial	Payments & Refunds	Payments	IMPLEMENTED
34	payments_paymentwebhooklog	Every inbound gateway callback, verified or not	Payments & Refunds	Payments	IMPLEMENTED
35	reviews_review	A verified-purchase review	Reviews & Ratings	Reviews	IMPLEMENTED

#	Table Name	Description	Module	Section	Status
36	reviews_reviewimage	Customer photographs attached to a review, with a generated thumbnail	Reviews & Ratings	Reviews	IMPLEMENTED
37	reviews_helpfulvote	One upvote by one customer on one review	Reviews & Ratings	Reviews	IMPLEMENTED
38	recommendations_recentlyviewed	Server-side browsing trail, owned by a user or a guest session under the same exclusive-or CHECK the cart uses	Recommendations & Personalisation	Recommendations	IMPLEMENTED
39	recommendations_productaffinity	Materialised co-purchase graph behind 'frequently bought together'	Recommendations & Personalisation	Recommendations	IMPLEMENTED
40	notifications_notification	One message to one person on one channel	Notifications	Notifications	IMPLEMENTED
41	notifications_notificationpreference	Per-customer opt-outs, one row per account	Notifications	Notifications	IMPLEMENTED
42	lkp_color	Colour master	Lookup & Reference Tables	Lookups	PROPOSED
43	lkp_size	Size master with a sort key	Lookup & Reference Tables	Lookups	PROPOSED
44	lkp_material	Fabric master, replacing the material enum on products_product	Lookup & Reference Tables	Lookups	PROPOSED
45	lkp_occasion	Occasion master (Casual, Formal, Wedding, Festive), replacing the product enum	Lookup & Reference Tables	Lookups	PROPOSED
46	lkp_gender	Gender/audience master, shared by products and customer profiles	Lookup & Reference Tables	Lookups	PROPOSED
47	lkp_age_group	Age banding (Adult, Teen, Kids 8–12, Infant)	Lookup & Reference Tables	Lookups	PROPOSED
48	lkp_season	Season master, replacing the season enum	Lookup & Reference Tables	Lookups	PROPOSED
49	lkp_pattern	Pattern master (Solid, Floral, Striped, Checked)	Lookup & Reference Tables	Lookups	PROPOSED
50	lkp_fit	Fit master (Slim, Regular, Relaxed, Oversized, Bodycon)	Lookup & Reference Tables	Lookups	PROPOSED
51	lkp_country	ISO country master for addresses, brand origin and product country of origin — all three are free-text VARCHAR(1...	Lookup & Reference Tables	Lookups	PROPOSED
52	warehouse	Physical fulfilment locations	Inventory & Warehousing	Inventory	PROPOSED
53	inventory_stock	Stock on hand per variant per warehouse	Inventory & Warehousing	Inventory	PROPOSED
54	inventory_transaction	Immutable ledger of every stock movement	Inventory & Warehousing	Inventory	PROPOSED
55	stock_alert	Open low-stock and out-of-stock signals, plus customer back-in-stock subscriptions	Inventory & Warehousing	Inventory	PROPOSED
56	product_price_history	Append-only price change log	Pricing, Offers & Promotions	Pricing	PROPOSED
57	discount_rule	Automatic discounts that need no coupon code — 'Buy 2 Get 1', 'Flat 30% on Denim', tiered cart discounts	Pricing, Offers & Promotions	Pricing	PROPOSED
58	offer	Customer-facing promotional banner copy for a discount rule — the marketing surface, separate from the pricing m...	Pricing, Offers & Promotions	Pricing	PROPOSED
59	flash_sale	Time-boxed sale event with a countdown	Pricing, Offers & Promotions	Pricing	PROPOSED
60	flash_sale_item	One product in a flash sale at an event-specific price and unit cap	Pricing, Offers & Promotions	Pricing	PROPOSED
61	delivery_partner	Courier master	Shipping, Returns & Exchanges	Shipping	PROPOSED
62	shipment_tracking_event	Scan-by-scan courier events	Shipping, Returns & Exchanges	Shipping	PROPOSED
63	return_request	A customer's request to send an item back	Shipping, Returns & Exchanges	Returns	PROPOSED
64	return_item	Which lines of an order are coming back, and how many units	Shipping, Returns & Exchanges	Returns	PROPOSED
65	exchange_request	A size or colour swap	Shipping, Returns & Exchanges	Returns	PROPOSED
66	homepage_banner	Scheduled promotional banners by placement	Content, Merchandising & Cms	Content	PROPOSED
67	hero_carousel_slide	Ordered slides for the homepage hero carousel, with per-slide dwell time	Content, Merchandising & Cms	Content	PROPOSED
68	product_video	Product videos and 360° spins	Content, Merchandising & Cms	Content	PROPOSED
69	variant_image	Images tied to a specific colourway	Content, Merchandising & Cms	Content	PROPOSED
70	lookbook	Editorial styling galleries — 'Wedding Season 2026', 'Office Capsule'	Content, Merchandising & Cms	Content	PROPOSED
71	lookbook_item	One shoppable hotspot on a lookbook image, positioned by percentage coordinates so it survives responsive scaling	Content, Merchandising & Cms	Content	PROPOSED
72	blog_category	Taxonomy for the fashion blog	Content, Merchandising & Cms	Content	PROPOSED

#	Table Name	Description	Module	Section	Status
73	blog_post	Fashion journal articles — SEO surface and top-of-funnel traffic	Content, Merchandising & Cms	Content	PROPOSED
74	influencer	Creator and brand-ambassador profiles behind lookbooks, articles and affiliate attribution	Content, Merchandising & Cms	Content	PROPOSED
75	influencer_campaign	A paid or affiliate collaboration with measurable attribution	Content, Merchandising & Cms	Content	PROPOSED
76	cms_page	Editable static pages — About, Privacy, Terms, Shipping Policy, Size Guide	Content, Merchandising & Cms	CMS	PROPOSED
77	faq_category	Grouping for FAQs (Orders, Returns, Payments, Sizing)	Content, Merchandising & Cms	CMS	PROPOSED
78	faq	Question and answer pairs for the help centre, with a helpfulness signal	Content, Merchandising & Cms	CMS	PROPOSED
79	product_question	Customer questions on a product page	Content, Merchandising & Cms	Q&A	PROPOSED
80	product_answer	An answer from a verified buyer, a seller or support	Content, Merchandising & Cms	Q&A	PROPOSED
81	loyalty_account	One points balance per customer with a tier	Customer Engagement & Loyalty	Loyalty	PROPOSED
82	loyalty_transaction	Append-only points ledger	Customer Engagement & Loyalty	Loyalty	PROPOSED
83	gift_card	Stored-value cards	Customer Engagement & Loyalty	Gift Cards	PROPOSED
84	gift_card_transaction	Every debit and credit against a card balance	Customer Engagement & Loyalty	Gift Cards	PROPOSED
85	referral	Refer-a-friend tracking	Customer Engagement & Loyalty	Referrals	PROPOSED
86	search_history	Every search query with its result count	Search, Behaviour & Ai Personalisation	Search	PROPOSED
87	trending_search	Aggregated popular queries per window	Search, Behaviour & Ai Personalisation	Search	PROPOSED
88	clickstream_event	Raw interaction events — the highest-volume table in the schema and the reason the partitioning section exists	Search, Behaviour & Ai Personalisation	Behaviour	PROPOSED
89	fashion_preference	A shopper's derived and declared style profile — the feature vector the recommendation engine reads	Search, Behaviour & Ai Personalisation	AI Personalisation	PROPOSED
90	recommendation_history	What was recommended, where, and whether it was clicked or bought	Search, Behaviour & Ai Personalisation	AI Personalisation	PROPOSED
91	notification_template	Database-backed notification copy	Content, Merchandising & Cms	Notifications	PROPOSED
92	email_template	HTML email layouts, separate from notification_template because an email has a wrapper, inline CSS and a plain-t...	Content, Merchandising & Cms	Notifications	PROPOSED
93	admin_role	Named back-office roles	Administration, Rbac & Audit	Admin	PROPOSED
94	admin_permission	Atomic permission keys, one per module and action	Administration, Rbac & Audit	Admin	PROPOSED
95	admin_role_permission	Bridge granting permissions to a role	Administration, Rbac & Audit	Admin	PROPOSED
96	admin_user_role	Bridge assigning back-office roles to staff accounts, with optional expiry for temporary elevation	Administration, Rbac & Audit	Admin	PROPOSED
97	audit_log	Row-level before/after change log for sensitive entities	Administration, Rbac & Audit	Audit	PROPOSED
98	activity_log	Customer-facing activity feed: logins, password changes, address edits, order placement	Administration, Rbac & Audit	Audit	PROPOSED
99	system_setting	Runtime configuration a business user may change without a deploy	Administration, Rbac & Audit	System	PROPOSED
100	contact_message	Contact-form submissions	Support & Communication	Support	PROPOSED
101	support_ticket	Customer support cases with SLA tracking	Support & Communication	Support	PROPOSED
102	support_message	One message in a ticket thread	Support & Communication	Support	PROPOSED
103	analytics_daily_sales	Pre-aggregated daily sales	Analytics Aggregation	Analytics	PROPOSED
104	analytics_product_daily	Per-product daily performance, and the funnel the current schema cannot produce: views to cart-adds to purchases	Analytics Aggregation	Analytics	PROPOSED
105	analytics_customer_daily	Customer cohort and retention metrics, computed once rather than on every dashboard load	Analytics Aggregation	Analytics	PROPOSED

DESIGN CONVENTIONS

Convention	Rule	Rationale
Primary keys	BIGSERIAL surrogate on every table	Monotonic integers keep B-tree inserts local. A random UUID primary key fragments every index on a table this size.
Public identifiers	uuid UUID DEFAULT gen_random_uuid(), UNIQUE	Exposed in URLs and API payloads so row counts and insert rates are not leaked. Secondary, never the PK.
Audit columns	created_at, updated_at TIMESTAMPTZ NOT NULL DEFAULT now()	Inherited by all 41 implemented tables from a shared abstract base; omitted from per-table grids only where noted.
Timestamps	TIMESTAMPTZ, stored UTC	The application runs Asia/Kolkata; storage is always UTC.
Money	NUMERIC(12,2) – NUMERIC(14,2) for aggregates	Never FLOAT. A rupee amount that cannot round-trip is a reconciliation failure.
Currency	CHAR(3), ISO 4217, default 'INR'	Stored per row, not assumed.
Enumerations	VARCHAR with a CHECK constraint	Not PostgreSQL ENUM types: altering an ENUM takes a lock, and adding a status should not require one. The lkp_* tables normalise the catalogue-facing ones.
Text	VARCHAR(n) where bounded, TEXT where not	Bounds are validation, documented per column.
Booleans	BOOLEAN NOT NULL with an explicit default	Never nullable — a three-valued flag is a bug waiting to be written.
Blank strings	' ' rather than NULL for optional short text	Django convention, carried into the schema. NULL is reserved for 'not applicable'.
Soft delete	Not used; is_active / status columns instead	Deletion is genuinely rare. PROTECT foreign keys guard the financial chain.
Naming	snake_case; tables prefixed by owning module	e.g. orders_order, payments_refund. Proposed tables adopt the same rule.
Constraint naming	pk_ / uq_ / fk_ / ck_ / idx_ prefixes	So a violation message names the rule that fired.
Cascade policy	CASCADE for owned children; PROTECT on financial edges; SET NULL where the child outlives the parent	Applied consistently — see the ERD notes.

AUTHENTICATION & USER MANAGEMENT

users_user [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
password	VARCHAR(128)	NOT NULL				
last_login	TIMESTAMPTZ	NULL				
is_superuser	BOOLEAN	NOT NULL	FALSE			
first_name	VARCHAR(150)	NOT NULL				
last_name	VARCHAR(150)	NOT NULL				
email	VARCHAR(254)	NOT NULL		UQ		
mobile_number	VARCHAR(16)	NOT NULL				
profile_image	VARCHAR(100)	NULL				
date_of_birth	DATE	NULL				
gender	VARCHAR(12)	NOT NULL				male female other undisclosed
is_email_verified	BOOLEAN	NOT NULL	FALSE			
is_mobile_verified	BOOLEAN	NOT NULL	FALSE			
is_active	BOOLEAN	NOT NULL	TRUE			
is_staff	BOOLEAN	NOT NULL	FALSE			
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; idx: user_created_at_idx
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained

Default ordering: created_at DESC · 17 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_users_user	id	Surrogate identity key.
UNIQUE	uq_users_user_email	email	Column-level uniqueness.
FOREIGN KEY	fk_users_user_groups	groups	REFERENCES auth_group(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_users_user_user_permissions	user_permissions	REFERENCES auth_permission(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	user_created_at_idx	created_at DESC	Composite B-tree.

Attribute	Detail
Purpose	Customer and staff accounts. Email is the login identifier — there is no separate username column — so the unique index on email is the credential lookup path for every sign-in. Passwords are stored as PBKDF2 hashes in the inherited password column; the application never stores plaintext.
Relationships	Parent of Address, Wishlist, Cart, Order, Review, HelpfulVote, CouponUsage, RecentlyViewed, Notification and NotificationPreference. Bridged to auth_group and auth_permission for Django RBAC.
Used By APIs	POST /auth/register · POST /auth/login · POST /auth/logout · POST /auth/token/refresh · POST /auth/token/verify · POST /auth/forgot-password · POST /auth/reset-password · POST /auth/change-password · GET PATCH /profile
Used By Frontend Pages	Sign In, Register, Forgot Password, Reset Password, My Account, Profile

users_address [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		

Column Name	Data Type	Nullable	Default	Key	References	Notes
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE; idx: address_user_default_idx
full_name	VARCHAR(150)	NOT NULL				
mobile	VARCHAR(16)	NOT NULL				
address_line_1	VARCHAR(255)	NOT NULL				
address_line_2	VARCHAR(255)	NOT NULL				
city	VARCHAR(100)	NOT NULL				
state	VARCHAR(100)	NOT NULL				
country	VARCHAR(100)	NOT NULL				
postal_code	VARCHAR(16)	NOT NULL				
is_default	BOOLEAN	NOT NULL	FALSE			idx: address_user_default_idx
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained

Default ordering: is_default DESC, created_at DESC · 13 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_users_address	id	Surrogate identity key.
UNIQUE	unique_default_address_per_user		COLUMNS ('user',) (('is_default', True))
FOREIGN KEY	fk_users_address_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	address_user_default_idx	user, is_default	Composite B-tree.

Attribute	Detail
Purpose	Delivery and billing addresses belonging to a customer. Exactly one address per customer may carry is_default = TRUE; that invariant is enforced by a partial unique index rather than by application code, so two concurrent 'make this default' requests cannot both win.
Relationships	Child of users_user (CASCADE). Read at checkout and snapshotted into orders_order.shipping_address as JSONB — the order keeps its own frozen copy, so editing an address never rewrites delivery history.
Used By APIs	GET POST /addresses · GET PATCH DELETE /addresses/{id}
Used By Frontend Pages	My Account → Addresses, Checkout → Delivery Address

users_user_groups [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Surrogate key
user_id	BIGINT	NOT NULL		FK/UQ*	users_user	ON DELETE CASCADE; composite UQ with the other side
group_id	BIGINT	NOT NULL		FK/UQ*	auth_group	ON DELETE CASCADE; composite UQ with the other side

Default ordering: None (unordered). · 3 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_users_user_groups	id	Surrogate identity key.
UNIQUE	uq_users_user_groups	user_id, group_id	One row per pair — the relationship itself carries no attributes.
FOREIGN KEY	fk_users_user_groups_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_users_user_groups_group_id	group_id	REFERENCES auth_group(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	idx_users_user_groups_group_id	group_id	Reverse traversal; the composite UQ already covers the forward direction.

Attribute	Detail
Purpose	Bridge assigning users to Django auth groups. Carries the role model today; the proposed admin_role / admin_role_permission tables supersede it for fine-grained back-office RBAC.
Relationships	Bridge: users_user ↔ auth_group. Composite unique on (user_id, group_id).
Used By APIs	Django admin only. Not exposed on the public API.
Used By Frontend Pages	Django Admin → Users

users_user_user_permissions [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Surrogate key
user_id	BIGINT	NOT NULL		FK/UQ*	users_user	ON DELETE CASCADE; composite UQ with the other side
permission_id	BIGINT	NOT NULL		FK/UQ*	auth_permission	ON DELETE CASCADE; composite UQ with the other side

Default ordering: None (unordered). · 3 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_users_user_user_permissions	id	Surrogate identity key.
UNIQUE	uq_users_user_user_permissions	user_id, permission_id	One row per pair — the relationship itself carries no attributes.
FOREIGN KEY	fk_users_user_user_permissions_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_users_user_user_permissions_permission_id	permission_id	REFERENCES auth_permission(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	idx_users_user_user_permissions_permission_id	permission_id	Reverse traversal; the composite UQ already covers the forward direction.

Attribute	Detail
Purpose	Bridge granting individual Django permissions directly to a user.
Relationships	Bridge: users_user ↔ auth_permission. Composite unique on (user_id, permission_id).
Used By APIs	Django admin only.
Used By Frontend Pages	Django Admin → Users

CATALOG TAXONOMY

catalog_category [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
slug	VARCHAR(255)	NOT NULL		UQ		
meta_title	VARCHAR(70)	NOT NULL				
meta_description	VARCHAR(170)	NOT NULL				
meta_keywords	VARCHAR(255)	NOT NULL				
name	VARCHAR(120)	NOT NULL		UQ		
description	TEXT	NOT NULL				
icon	VARCHAR(100)	NULL				
image	VARCHAR(100)	NULL				
banner_image	VARCHAR(100)	NULL				
menu_image	VARCHAR(100)	NULL				
is_active	BOOLEAN	NOT NULL	TRUE			idx: category_active_order_idx, category_featured_idx
is_featured	BOOLEAN	NOT NULL	FALSE			idx: category_featured_idx
is_trending	BOOLEAN	NOT NULL	FALSE			B-tree index
is_luxury	BOOLEAN	NOT NULL	FALSE			B-tree index
display_order	SMALLINT	NOT NULL	0			idx: category_active_order_idx

Default ordering: display_order ASC, name ASC · 19 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_catalog_category	id	Surrogate identity key.
UNIQUE	uq_catalog_category_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_catalog_category_slug	slug	Column-level uniqueness.
UNIQUE	uq_catalog_category_name	name	Column-level uniqueness.
INDEX	category_active_order_idx	is_active, display_order	Composite B-tree.
INDEX	category_featured_idx	is_featured, is_active	Composite B-tree.

Attribute	Detail
Purpose	Top level of the merchandising tree (Women, Men, Kids, Accessories...). Carries its own SEO block and four separate image slots because the same category is rendered as a nav tile, a mega-menu panel, a landing banner and a homepage card, each at a different aspect ratio.
Relationships	Parent of catalog_subcategory and products_product (PROTECT — a category with live products cannot be deleted). Bridged to Brand, Collection and Coupon.
Used By APIs	GET /categories · /categories/tree · /categories/mega-menu · /categories/homepage · /categories/featured · /categories/trending · /categories/luxury · /categories/{slug} · /categories/{slug}/subcategories · /categories/{slug}/brands · /categories/{slug}/collections
Used By Frontend Pages	Header Mega Menu, Homepage Category Strip, Category Landing, Breadcrumbs

catalog_subcategory [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
slug	VARCHAR(255)	NOT NULL		UQ		
meta_title	VARCHAR(70)	NOT NULL				
meta_description	VARCHAR(170)	NOT NULL				
meta_keywords	VARCHAR(255)	NOT NULL				
category_id	BIGINT	NOT NULL		FK	catalog_category	ON DELETE CASCADE; idx: subcat_cat_active_order_idx
name	VARCHAR(120)	NOT NULL				
description	TEXT	NOT NULL				
image	VARCHAR(100)	NULL				
banner_image	VARCHAR(100)	NULL				
is_active	BOOLEAN	NOT NULL	TRUE			idx: subcat_cat_active_order_idx
display_order	SMALLINT	NOT NULL	0			idx: subcat_cat_active_order_idx

Default ordering: category__display_order ASC, display_order ASC, name ASC · 15 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_catalog_subcategory	id	Surrogate identity key.
UNIQUE	uq_catalog_subcategory_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_catalog_subcategory_slug	slug	Column-level uniqueness.
UNIQUE	unique_subcategory_name_per_category		COLUMNS ('category', 'name')
FOREIGN KEY	fk_catalog_subcategory_category_id	category_id	REFERENCES catalog_category(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	subcat_cat_active_order_idx	category, is_active, display_order	Composite B-tree.

Attribute	Detail
Purpose	Second level of the tree (Dresses, Shirts, Sarees...). This is the unit the recommendation engine treats as a neighbourhood: 'Dresses' is a useful related-products scope, 'Women' is the whole store.
Relationships	Child of catalog_category (CASCADE). Parent of products_product (PROTECT). Name is unique per parent category, not globally — 'Shirts' may exist under both Men and Women.
Used By APIs	GET /subcategories · /subcategories/{slug} · /categories/{slug}/subcategories
Used By Frontend Pages	Mega Menu columns, Category Landing filter rail, Product Listing

catalog_brand [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
slug	VARCHAR(255)	NOT NULL		UQ		

Column Name	Data Type	Nullable	Default	Key	References	Notes
meta_title	VARCHAR(70)	NOT NULL				
meta_description	VARCHAR(170)	NOT NULL				
meta_keywords	VARCHAR(255)	NOT NULL				
name	VARCHAR(120)	NOT NULL		UQ		
description	TEXT	NOT NULL				
logo	VARCHAR(100)	NULL				
banner	VARCHAR(100)	NULL				
website	VARCHAR(255)	NOT NULL				
country	VARCHAR(100)	NOT NULL				
founded_year	SMALLINT	NULL				
is_active	BOOLEAN	NOT NULL	TRUE			idx: brand_active_order_idx, brand_popularity_idx
is_featured	BOOLEAN	NOT NULL	FALSE			B-tree index
is_luxury	BOOLEAN	NOT NULL	FALSE			idx: brand_luxury_idx
display_order	SMALLINT	NOT NULL	0			idx: brand_active_order_idx
popularity_score	INTEGER	NOT NULL	0			idx: brand_popularity_idx

Default ordering: display_order ASC, name ASC · 20 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_catalog_brand	id	Surrogate identity key.
UNIQUE	uq_catalog_brand_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_catalog_brand_slug	slug	Column-level uniqueness.
UNIQUE	uq_catalog_brand_name	name	Column-level uniqueness.
FOREIGN KEY	fk_catalog_brand_categories	categories	REFERENCES catalog_category(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	brand_active_order_idx	is_active, display_order	Composite B-tree.
INDEX	brand_popularity_idx	is_active, popularity_score DESC	Composite B-tree.
INDEX	brand_luxury_idx	is_luxury, is_active	Composite B-tree.

Attribute	Detail
Purpose	Brand master. popularity_score is a denormalised ranking input maintained by the application so the brand rails sort without joining orders.
Relationships	Parent of products_product (PROTECT). Bridged many-to-many to Category via catalog_brand_categories, and to Coupon via coupons_coupon_brands.
Used By APIs	GET /brands · /brands/featured · /brands/luxury · /brands/popular · /brands/top · /brands/{slug} · /categories/{slug}/brands
Used By Frontend Pages	Brand Directory, Brand Landing, Homepage Featured Brands, Filter rail

catalog_brand_categories [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Surrogate key
brand_id	BIGINT	NOT NULL		FK/UQ*	catalog_brand	ON DELETE CASCADE; composite UQ with the other side
category_id	BIGINT	NOT NULL		FK/UQ*	catalog_category	ON DELETE CASCADE; composite UQ with the other side

Default ordering: None (unordered). · 3 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_catalog_brand_categories	id	Surrogate identity key.
UNIQUE	uq_catalog_brand_categories	brand_id, category_id	One row per pair — the relationship itself carries no attributes.
FOREIGN KEY	fk_catalog_brand_categories_brand_id	brand_id	REFERENCES catalog_brand(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_catalog_brand_categories_category_id	category_id	REFERENCES catalog_category(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	idx_catalog_brand_categories_category_id	category_id	Reverse traversal; the composite UQ already covers the forward direction.

Attribute	Detail
Purpose	Bridge: which categories a brand sells into. Drives the per-category brand rail.
Relationships	Bridge: catalog_brand ↔ catalog_category. Composite unique on (brand_id, category_id).
Used By APIs	GET /categories/{slug}/brands
Used By Frontend Pages	Category Landing → Shop by Brand

catalog_collection [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
slug	VARCHAR(255)	NOT NULL		UQ		
meta_title	VARCHAR(70)	NOT NULL				
meta_description	VARCHAR(170)	NOT NULL				
meta_keywords	VARCHAR(255)	NOT NULL				
title	VARCHAR(150)	NOT NULL		UQ		
description	TEXT	NOT NULL				
type	VARCHAR(24)	NOT NULL	new_arrivals			new_arrivals trending luxury editors_picks best_sellers festival wedding su...; idx: collection_type_idx
image	VARCHAR(100)	NULL				
banner	VARCHAR(100)	NULL				
is_active	BOOLEAN	NOT NULL	TRUE			idx: collection_active_order_idx, collection_type_idx
is_featured	BOOLEAN	NOT NULL	FALSE			B-tree index
display_order	SMALLINT	NOT NULL	0			idx: collection_active_order_idx

Default ordering: display_order ASC, title ASC · 16 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_catalog_collection	id	Surrogate identity key.
UNIQUE	uq_catalog_collection_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_catalog_collection_slug	slug	Column-level uniqueness.
UNIQUE	uq_catalog_collection_title	title	Column-level uniqueness.
FOREIGN KEY	fk_catalog_collection_categories	categories	REFERENCES catalog_category(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	collection_active_order_idx	is_active, display_order	Composite B-tree.

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
INDEX	collection_type_idx	type, is_active	Composite B-tree.

Attribute	Detail
Purpose	Curated merchandising groupings — New Arrivals, Trending, Luxury, Editor's Picks, Best Sellers, Seasonal. The type column classifies the rail so the homepage can request one without hard-coding slugs.
Relationships	Referenced by products_product.collection_id (SET NULL). Bridged to Category via catalog_collection_categories.
Used By APIs	GET /collections · /collections/featured · /collections/homepage · /collections/editors-picks · /collections/seasonal · /collections/{slug}
Used By Frontend Pages	Homepage rails, Collection Landing, Seasonal Campaign pages

catalog_collection_categories [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Surrogate key
collection_id	BIGINT	NOT NULL		FK/UQ*	catalog_collection	ON DELETE CASCADE; composite UQ with the other side
category_id	BIGINT	NOT NULL		FK/UQ*	catalog_category	ON DELETE CASCADE; composite UQ with the other side

Default ordering: None (unordered). · 3 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_catalog_collection_categories	id	Surrogate identity key.
UNIQUE	uq_catalog_collection_categories	collection_id, category_id	One row per pair — the relationship itself carries no attributes.
FOREIGN KEY	fk_catalog_collection_categories_collection_id	collection_id	REFERENCES catalog_collection(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_catalog_collection_categories_category_id	category_id	REFERENCES catalog_category(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	idx_catalog_collection_categories_category_id	category_id	Reverse traversal; the composite UQ already covers the forward direction.

Attribute	Detail
Purpose	Bridge scoping a collection to one or more categories.
Relationships	Bridge: catalog_collection ↔ catalog_category.
Used By APIs	GET /categories/{slug}/collections
Used By Frontend Pages	Category Landing → Collections strip

PRODUCT CATALOG

products_product [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
slug	VARCHAR(255)	NOT NULL		UQ		
meta_title	VARCHAR(70)	NOT NULL				
meta_description	VARCHAR(170)	NOT NULL				
meta_keywords	VARCHAR(255)	NOT NULL				
name	VARCHAR(200)	NOT NULL				B-tree index
short_description	VARCHAR(300)	NOT NULL				
long_description	TEXT	NOT NULL				
sku	VARCHAR(50)	NOT NULL		UQ		
barcode	VARCHAR(14)	NOT NULL				
category_id	BIGINT	NOT NULL		FK	catalog_category	ON DELETE PROTECT; idx: product_cat_listing_idx
subcategory_id	BIGINT	NOT NULL		FK	catalog_subcategory	ON DELETE PROTECT; idx: product_subcat_listing_idx
brand_id	BIGINT	NOT NULL		FK	catalog_brand	ON DELETE PROTECT; idx: product_brand_listing_idx
collection_id	BIGINT	NULL		FK	catalog_collection	ON DELETE SET NULL; B-tree index
mrp	NUMERIC(12,2)	NOT NULL				
selling_price	NUMERIC(12,2)	NOT NULL				idx: product_price_idx
discount_percentage	NUMERIC(5,2)	NOT NULL	0.00			idx: product_discount_idx
discount_amount	NUMERIC(12,2)	NOT NULL	0.00			
tax_percentage	NUMERIC(5,2)	NOT NULL	5.00			
currency	VARCHAR(3)	NOT NULL	INR			INR USD EUR GBP AED
gender	VARCHAR(12)	NOT NULL	unisex			male female unisex other undisclosed; idx: product_gender_idx
material	VARCHAR(16)	NOT NULL				cotton linen silk wool denim leather polyester rayon viscose georgette ...; B-tree index
fit	VARCHAR(12)	NOT NULL				slim regular relaxed oversized bodycon a_line straight tapered
sleeve_type	VARCHAR(16)	NOT NULL				sleeveless cap short three_quarter long puff bell na
neck_type	VARCHAR(16)	NOT NULL				round v_neck collared boat square halter mandarin sweetheart na
pattern	VARCHAR(16)	NOT NULL				solid printed floral striped checked embroidered colourblock animal abstr...
occasion	VARCHAR(12)	NOT NULL				casual formal party wedding festive sports lounge beach work; B-tree index
season	VARCHAR(12)	NOT NULL	all_season			summer winter monsoon spring autumn all_season; B-tree index
country_of_origin	VARCHAR(100)	NOT NULL				
weight_grams	INTEGER	NULL				
dimensions	VARCHAR(60)	NOT NULL				

Column Name	Data Type	Nullable	Default	Key	References	Notes
warranty	VARCHAR(150)	NOT NULL				
care_instructions	TEXT	NOT NULL				
return_policy	VARCHAR(200)	NOT NULL				
shipping_info	VARCHAR(200)	NOT NULL				
estimated_delivery_days	SMALLINT	NOT NULL	5			
rating_average	NUMERIC(3,2)	NOT NULL	0.00			idx: product_rating_idx
rating_count	INTEGER	NOT NULL	0			
review_count	INTEGER	NOT NULL	0			
wishlist_count	INTEGER	NOT NULL	0			
view_count	INTEGER	NOT NULL	0			B-tree index
purchase_count	INTEGER	NOT NULL	0			idx: product_bestseller_idx
total_stock	INTEGER	NOT NULL	0			B-tree index
stock_status	VARCHAR(16)	NOT NULL	out_of_stock			in_stock low_stock out_of_stock discontinued; B-tree index
is_active	BOOLEAN	NOT NULL	TRUE			idx: product_visible_new_idx, product_cat_listing_idx
is_featured	BOOLEAN	NOT NULL	FALSE			idx: product_featured_idx
is_new_arrival	BOOLEAN	NOT NULL	FALSE			idx: product_newarrival_idx
is_best_seller	BOOLEAN	NOT NULL	FALSE			B-tree index
is_trending	BOOLEAN	NOT NULL	FALSE			idx: product_trending_idx
is_luxury	BOOLEAN	NOT NULL	FALSE			idx: product_luxury_idx
is_recommended	BOOLEAN	NOT NULL	FALSE			B-tree index
is_returnable	BOOLEAN	NOT NULL	TRUE			
published_at	TIMESTAMPTZ	NULL				idx: product_visible_new_idx, product_cat_listing_idx

Default ordering: published_at DESC, created_at DESC · 56 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_products_product	id	Surrogate identity key.
UNIQUE	uq_products_product_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_products_product_slug	slug	Column-level uniqueness.
UNIQUE	uq_products_product_sku	sku	Column-level uniqueness.
CHECK	product_selling_price_not_above_mrp		((selling_price <= ' , F(mrp)))
CHECK	product_mrp_not_negative		((mrp >= ' , Decimal('0')))
FOREIGN KEY	fk_products_product_category_id	category_id	REFERENCES catalog_category(id) ON DELETE PROTECT ON UPDATE CASCADE
FOREIGN KEY	fk_products_product_subcategory_id	subcategory_id	REFERENCES catalog_subcategory(id) ON DELETE PROTECT ON UPDATE CASCADE
FOREIGN KEY	fk_products_product_brand_id	brand_id	REFERENCES catalog_brand(id) ON DELETE PROTECT ON UPDATE CASCADE
FOREIGN KEY	fk_products_product_collection_id	collection_id	REFERENCES catalog_collection(id) ON DELETE SET NULL ON UPDATE CASCADE
FOREIGN KEY	fk_products_product_tags	tags	REFERENCES products_producttag(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	product_visible_new_idx	is_active, published_at DESC	Composite B-tree.
INDEX	product_cat_listing_idx	category, is_active, published_at DESC	Composite B-tree.
INDEX	product_subcat_listing_idx	subcategory, is_active, published_at DESC	Composite B-tree.

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
INDEX	product_brand_listing_idx	brand, is_active, published_at DESC	Composite B-tree.
INDEX	product_price_idx	is_active, selling_price	Composite B-tree.
INDEX	product_discount_idx	is_active, discount_percentage DESC	Composite B-tree.
INDEX	product_rating_idx	is_active, rating_average DESC	Composite B-tree.
INDEX	product_bestseller_idx	is_active, purchase_count DESC	Composite B-tree.
INDEX	product_featured_idx	is_featured, is_active	Composite B-tree.
INDEX	product_trending_idx	is_trending, is_active	Composite B-tree.
INDEX	product_newarrival_idx	is_new_arrival, is_active	Composite B-tree.
INDEX	product_luxury_idx	is_luxury, is_active	Composite B-tree.
INDEX	product_gender_idx	gender, is_active	Composite B-tree.

Attribute	Detail
Purpose	The master product record and the widest table in the schema at 57 columns. Holds identity, taxonomy, pricing, thirteen fashion attributes, logistics copy, six denormalised engagement counters and eight merchandising flags. The counters (rating_average, view_count, purchase_count, wishlist_count...) are maintained by the application with F() expressions so that listing pages sort and filter without aggregating child tables on every request.
Relationships	Child of Category, SubCategory and Brand (all PROTECT) and optionally Collection (SET NULL). Parent of ProductImage, ProductVariant, ProductAttribute, ProductSpecification, WishlistItem, CartItem, Review, RecentlyViewed and ProductAffinity. Referenced by OrderItem with SET NULL, because an order line must outlive the product it sold.
Used By APIs	GET /products · /products/{slug} · /products/{slug}/quick-view · /products/{slug}/availability · /products/{slug}/related · /products/{slug}/similar · /products/search · /products/suggestions · /products/filters · /products/homepage · /products/featured · /products/trending · /products/new-arrivals · /products/best-sellers · /products/luxury · /products/flash-sale · /products/recommended · /products/editors-picks · /products/low-stock
Used By Frontend Pages	Homepage, Product Listing, Product Detail, Search Results, Quick View, Wishlist, Cart, Admin Catalogue

products_producttag [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
slug	VARCHAR(255)	NOT NULL		UQ		
meta_title	VARCHAR(70)	NOT NULL				
meta_description	VARCHAR(170)	NOT NULL				
meta_keywords	VARCHAR(255)	NOT NULL				
name	VARCHAR(80)	NOT NULL		UQ		
description	TEXT	NOT NULL				
is_active	BOOLEAN	NOT NULL	TRUE			B-tree index

Default ordering: name ASC · 11 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_products_producttag	id	Surrogate identity key.
UNIQUE	uq_products_producttag_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_products_producttag_slug	slug	Column-level uniqueness.
UNIQUE	uq_products_producttag_name	name	Column-level uniqueness.

Attribute	Detail
Purpose	Free-form merchandising labels (Festive, Office Wear, Monsoon Ready).
Relationships	Bridged many-to-many to Product via products_product_tags.
Used By APIs	GET /product-tags · /product-tags/{slug} · /product-tags/{slug}/products
Used By Frontend Pages	Product Detail → tag chips, Tag Landing

products_product_tags [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Surrogate key
product_id	BIGINT	NOT NULL		FK/UQ*	products_product	ON DELETE CASCADE; composite UQ with the other side
producttag_id	BIGINT	NOT NULL		FK/UQ*	products_producttag	ON DELETE CASCADE; composite UQ with the other side

Default ordering: None (unordered). · 3 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_products_product_tags	id	Surrogate identity key.
UNIQUE	uq_products_product_tags	product_id, producttag_id	One row per pair — the relationship itself carries no attributes.
FOREIGN KEY	fk_products_product_tags_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_products_product_tags_producttag_id	producttag_id	REFERENCES products_producttag(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	idx_products_product_tags_producttag_id	producttag_id	Reverse traversal; the composite UQ already covers the forward direction.

Attribute	Detail
Purpose	Bridge assigning merchandising tags to products.
Relationships	Bridge: products_product ↔ products_producttag.
Used By APIs	GET /product-tags/{slug}/products
Used By Frontend Pages	Tag Landing

products_productimage [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; idx: image_product_order_idx
image	VARCHAR(100)	NOT NULL				
thumbnail	VARCHAR(100)	NULL				
alt_text	VARCHAR(200)	NOT NULL				
display_order	SMALLINT	NOT NULL	0			idx: image_product_order_idx
is_primary	BOOLEAN	NOT NULL	FALSE			B-tree index

Default ordering: is_primary DESC, display_order ASC, id ASC · 10 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_products_productimage	id	Surrogate identity key.
UNIQUE	uq_products_productimage_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_primary_image_per_product		COLUMNS ('product',) (('is_primary', True))
FOREIGN KEY	fk_products_productimage_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	image_product_order_idx	product, display_order	Composite B-tree.

Attribute	Detail
Purpose	Product gallery. At most one row per product may be primary, enforced by a partial unique index — the card image is a single-valued fact and must not depend on which row the query happens to return first.
Relationships	Child of products_product (CASCADE).
Used By APIs	Embedded in /products/{slug} and every card payload.
Used By Frontend Pages	Product Detail gallery, Listing cards, Quick View, Cart line thumbnails

products_productvariant [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; idx: variant_product_idx
sku	VARCHAR(50)	NOT NULL		UQ		
barcode	VARCHAR(14)	NOT NULL				
color	VARCHAR(40)	NOT NULL				idx: variant_color_idx
color_code	VARCHAR(7)	NOT NULL				
size	VARCHAR(8)	NOT NULL				XS S M L XL XXL 3XL FREE UK6 UK7 UK8 UK9 UK10 UK11; idx: variant_size_idx
stock	INTEGER	NOT NULL	0			
reserved_stock	INTEGER	NOT NULL	0			
price_override	NUMERIC(12,2)	NULL				
image_override	VARCHAR(100)	NULL				
weight_grams	INTEGER	NULL				
is_active	BOOLEAN	NOT NULL	TRUE			idx: variant_product_idx

Default ordering: product ASC, color ASC, size ASC · 16 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_products_productvariant	id	Surrogate identity key.
UNIQUE	uq_products_productvariant_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_products_productvariant_sku	sku	Column-level uniqueness.
UNIQUE	unique_variant_per_product		COLUMNS ('product', 'color', 'size')
CHECK	variant_reserved_not_above_stock		((reserved_stock <= ', F(stock)))
FOREIGN KEY	fk_products_productvariant_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
INDEX	variant_product_idx	product, is_active	Composite B-tree.
INDEX	variant_color_idx	color	Composite B-tree.
INDEX	variant_size_idx	size	Composite B-tree.

Attribute	Detail
Purpose	The sellable unit: one row per colour and size. Holds the only authoritative stock figure in the schema. reserved_stock is held between checkout and payment capture and is constrained never to exceed stock, so the database itself refuses to oversell.
Relationships	Child of products_product (CASCADE). Referenced by CartItem (CASCADE) and OrderItem (SET NULL). Unique on (product, colour, size).
Used By APIs	GET /products/{slug}/availability · embedded in /products/{slug}
Used By Frontend Pages	Product Detail size/colour picker, Cart, Checkout, Admin Inventory

products_productattribute [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; B-tree index
key	VARCHAR(80)	NOT NULL				
value	VARCHAR(255)	NOT NULL				

Default ordering: key ASC · 7 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_products_productattribute	id	Surrogate identity key.
UNIQUE	uq_products_productattribute_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_attribute_key_per_product		COLUMNS ('product', 'key')
FOREIGN KEY	fk_products_productattribute_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE

Attribute	Detail
Purpose	Arbitrary key/value facts for filtering. One value per key per product.
Relationships	Child of products_product (CASCADE). Unique on (product, key).
Used By APIs	Embedded in /products/{slug}; feeds /products/filters
Used By Frontend Pages	Product Detail → Attributes, Listing filter rail

products_productspecification [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		

Column Name	Data Type	Nullable	Default	Key	References	Notes
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; B-tree index
label	VARCHAR(80)	NOT NULL				
value	VARCHAR(255)	NOT NULL				
display_order	SMALLINT	NOT NULL	0			

Default ordering: display_order ASC, label ASC · 8 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_products_productspecification	id	Surrogate identity key.
UNIQUE	uq_products_productspecification_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_specification_label_per_product		COLUMNS ('product', 'label')
FOREIGN KEY	fk_products_productspecification_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE

Attribute	Detail
Purpose	Ordered display-only specification rows (Fabric, Fit, Wash Care).
Relationships	Child of products_product (CASCADE). Unique on (product, label).
Used By APIs	Embedded in /products/{slug}
Used By Frontend Pages	Product Detail → Specifications table

WISHLIST & CART

wishlist_wishlist [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
user_id	BIGINT	NOT NULL		FK/UQ	users_user	ON DELETE CASCADE; one-to-one

Default ordering: created_at DESC · 5 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_wishlist_wishlist	id	Surrogate identity key.
UNIQUE	uq_wishlist_wishlist_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_wishlist_wishlist_user_id	user_id	Column-level uniqueness.
FOREIGN KEY	fk_wishlist_wishlist_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE

Attribute	Detail
Purpose	One wishlist per signed-in customer. A singleton, enforced by a one-to-one key.
Relationships	One-to-one with users_user (CASCADE). Parent of wishlist_wishlistitem.
Used By APIs	GET /wishlist · /wishlist/count · /wishlist/slugs · POST /wishlist/clear
Used By Frontend Pages	Wishlist page, header heart badge

wishlist_wishlistitem [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; idx: wishlistitem_recent_idx
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
wishlist_id	BIGINT	NOT NULL		FK	wishlist_wishlist	ON DELETE CASCADE; idx: wishlistitem_recent_idx
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; B-tree index

Default ordering: created_at DESC · 6 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_wishlist_wishlistitem	id	Surrogate identity key.
UNIQUE	uq_wishlist_wishlistitem_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_product_per_wishlist		COLUMNS ('wishlist', 'product')
FOREIGN KEY	fk_wishlist_wishlistitem_wishlist_id	wishlist_id	REFERENCES wishlist_wishlist(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_wishlist_wishlistitem_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	wishlistitem_recent_idx	wishlist, created_at DESC	Composite B-tree.

Attribute	Detail
Purpose	One saved product. Deliberately holds no variant: a shopper hearts 'this dress', not 'this dress in navy, size M'. The size decision happens when the item moves to the cart.
Relationships	Child of Wishlist and Product (both CASCADE). Unique on (wishlist, product).
Used By APIs	POST /wishlist/add · /wishlist/remove · /wishlist/toggle · /wishlist/move-to-cart
Used By Frontend Pages	Wishlist page, Product Detail heart, Listing card heart

cart_cart [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained; idx: cart_abandoned_idx
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
user_id	BIGINT	NULL		FK	users_user	ON DELETE CASCADE; idx: cart_user_active_idx
session_key	VARCHAR(64)	NOT NULL				idx: cart_session_active_idx
is_active	BOOLEAN	NOT NULL	TRUE			idx: cart_user_active_idx, cart_session_active_idx
coupon_code	VARCHAR(40)	NOT NULL				
coupon_discount	NUMERIC(12,2)	NOT NULL	0.00			
currency	VARCHAR(3)	NOT NULL	INR			INR USD EUR GBP AED

Default ordering: updated_at DESC · 10 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_cart_cart	id	Surrogate identity key.
UNIQUE	uq_cart_cart_uuid	uuid	Column-level uniqueness.
CHECK	cart_owned_by_user_xor_session		((('session_key' > ''), ('user IS NULL ', False)), (('session_key > ''), ('user IS NULL ', True)))
UNIQUE	unique_active_cart_per_user		COLUMNS ('user',) (('is_active', True), ('user IS NULL ', False))
UNIQUE	unique_active_cart_per_session		COLUMNS ('session_key',) (('is_active', True), ('user IS NULL ', True))
FOREIGN KEY	fk_cart_cart_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	cart_user_active_idx	user, is_active	Composite B-tree.
INDEX	cart_session_active_idx	session_key, is_active	Composite B-tree.
INDEX	cart_abandoned_idx	is_active, updated_at DESC	Composite B-tree.

Attribute	Detail
Purpose	A shopping bag owned by either a signed-in customer or an anonymous session, never both and never neither — a CHECK constraint enforces the exclusive-or. Two partial unique indexes guarantee one active cart per user and one per guest session.
Relationships	Optional child of users_user (CASCADE). Parent of cart_cartitem. Consumed and deactivated by checkout.
Used By APIs	GET /cart · /cart/summary · /cart/count · POST /cart/add · /cart/update · /cart/remove · /cart/increase · /cart/decrease · /cart/clear · /cart/merge · /cart/apply-coupon · /cart/remove-coupon · /cart/checkout
Used By Frontend Pages	Cart page, Mini-cart drawer, header bag badge, Checkout

cart_cartitem [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		

Column Name	Data Type	Nullable	Default	Key	References	Notes
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
cart_id	BIGINT	NOT NULL		FK	cart_cart	ON DELETE CASCADE; idx: cartitem_cart_saved_idx
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; B-tree index
variant_id	BIGINT	NOT NULL		FK	products_productvariant	ON DELETE CASCADE; B-tree index
quantity	SMALLINT	NOT NULL	1			
saved_for_later	BOOLEAN	NOT NULL	FALSE			idx: cartitem_cart_saved_idx
unit_price	NUMERIC(12,2)	NOT NULL	0.00			
unit_mrp	NUMERIC(12,2)	NOT NULL	0.00			
discount	NUMERIC(12,2)	NOT NULL	0.00			
tax	NUMERIC(12,2)	NOT NULL	0.00			
subtotal	NUMERIC(12,2)	NOT NULL	0.00			
total	NUMERIC(12,2)	NOT NULL	0.00			

Default ordering: saved_for_later ASC, created_at DESC · 15 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_cart_cartitem	id	Surrogate identity key.
UNIQUE	uq_cart_cartitem_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_variant_per_cart		COLUMNS ('cart', 'variant')
CHECK	cartitem_quantity_at_least_one		((quantity >= '1'))
CHECK	cartitem_quantity_within_limit		((quantity <= '10'))
FOREIGN KEY	fk_cart_cartitem_cart_id	cart_id	REFERENCES cart_cart(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_cart_cartitem_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_cart_cartitem_variant_id	variant_id	REFERENCES products_productvariant(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	cartitem_cart_saved_idx	cart, saved_for_later	Composite B-tree.

Attribute	Detail
Purpose	One bag line. Money columns are recomputed on every mutation rather than at read time, so the cart total is a stored fact the checkout can trust. saved_for_later moves a line out of the payable set without deleting it. Quantity is bounded 1..10 by CHECK constraints.
Relationships	Child of Cart, Product and ProductVariant (all CASCADE). Unique on (cart, variant).
Used By APIs	POST /cart/save-for-later · /cart/move-to-bag · /cart/move-to-wishlist
Used By Frontend Pages	Cart page, Saved For Later section, Mini-cart

ORDERS & FULFILMENT

orders_order [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; idx: order_user_recent_idx, order_status_idx
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
order_number	VARCHAR(32)	NOT NULL		UQ		
invoice_number	VARCHAR(32)	NOT NULL				B-tree index
invoice_file	VARCHAR(100)	NULL				
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE PROTECT; idx: order_user_recent_idx
shipping_address	JSONB	NOT NULL				
billing_address	JSONB	NOT NULL				
subtotal	NUMERIC(12,2)	NOT NULL				
discount	NUMERIC(12,2)	NOT NULL	0.00			
coupon_code	VARCHAR(40)	NOT NULL				
coupon_discount	NUMERIC(12,2)	NOT NULL	0.00			
shipping_charge	NUMERIC(12,2)	NOT NULL	0.00			
platform_fee	NUMERIC(12,2)	NOT NULL	0.00			
tax	NUMERIC(12,2)	NOT NULL	0.00			
grand_total	NUMERIC(12,2)	NOT NULL				
currency	VARCHAR(3)	NOT NULL	INR			INR USD EUR GBP AED
payment_method	VARCHAR(16)	NOT NULL	cod			card upi net_banking wallet cod
payment_status	VARCHAR(20)	NOT NULL	pending			pending authorised paid failed cancelled partially_refunded refunded; idx: order_payment_idx
payment_reference	VARCHAR(100)	NOT NULL				
status	VARCHAR(20)	NOT NULL	pending			pending confirmed processing packed shipped out_for_delivery delivered canc...; idx: order_status_idx
delivery_status	VARCHAR(20)	NOT NULL	not_dispatched			not_dispatched dispatched in_transit out_for_delivery delivered failed rto; idx: order_delivery_idx
delivery_method	VARCHAR(16)	NOT NULL	standard			standard express scheduled
estimated_delivery_date	DATE	NULL				
delivered_at	TIMESTAMP TZ	NULL				
cancelled_at	TIMESTAMP TZ	NULL				
cancel_reason	VARCHAR(255)	NOT NULL				
notes	TEXT	NOT NULL				
internal_notes	TEXT	NOT NULL				
stock_committed	BOOLEAN	NOT NULL	FALSE			

Default ordering: created_at DESC · 32 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_orders_order	id	Surrogate identity key.
UNIQUE	uq_orders_order_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_orders_order_order_number	order_number	Column-level uniqueness.
CHECK	order_grand_total_not_negative		((grand_total >= ', Decimal('0')))
CHECK	order_subtotal_not_negative		((subtotal >= ', Decimal('0')))
FOREIGN KEY	fk_orders_order_user_id	user_id	REFERENCES users_user(id) ON DELETE PROTECT ON UPDATE CASCADE
INDEX	order_user_recent_idx	user, created_at DESC	Composite B-tree.
INDEX	order_status_idx	status, created_at DESC	Composite B-tree.
INDEX	order_payment_idx	payment_status, created_at DESC	Composite B-tree.
INDEX	order_delivery_idx	delivery_status, created_at DESC	Composite B-tree.

Attribute	Detail
Purpose	The order header and the financial record of a sale. Addresses are stored as JSONB snapshots, not foreign keys: the delivery address on a two-year-old order must remain what it was on the day, whatever the customer has since edited. Every money component is stored separately so finance can reconcile subtotal, discount, coupon, shipping, platform fee and tax against the total.
Relationships	Child of users_user (PROTECT — a customer with orders cannot be hard-deleted). Parent of OrderItem, OrderStatusHistory, Shipment and Payment.
Used By APIs	GET /orders · /orders/{order_number} · /orders/{order_number}/tracking · /orders/{order_number}/shipments · /orders/{order_number}/invoice · POST /checkout/place-order · /orders/{order_number}/cancel · /orders/{order_number}/reorder · /orders/{order_number}/request-return · /orders/{order_number}/update-status
Used By Frontend Pages	My Orders, Order Detail, Order Tracking, Order Confirmation, Admin Orders

orders_orderitem [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
order_id	BIGINT	NOT NULL		FK	orders_order	ON DELETE CASCADE; idx: orderitem_order_idx
product_id	BIGINT	NULL		FK	products_product	ON DELETE SET NULL; B-tree index
variant_id	BIGINT	NULL		FK	products_productvariant	ON DELETE SET NULL; B-tree index
product_name	VARCHAR(200)	NOT NULL				
product_slug	VARCHAR(255)	NOT NULL				B-tree index
brand_name	VARCHAR(120)	NOT NULL				
sku	VARCHAR(50)	NOT NULL				idx: orderitem_sku_idx
size	VARCHAR(16)	NOT NULL				
color	VARCHAR(40)	NOT NULL				
image_url	VARCHAR(500)	NOT NULL				
mrp	NUMERIC(12,2)	NOT NULL				
selling_price	NUMERIC(12,2)	NOT NULL				
discount	NUMERIC(12,2)	NOT NULL	0.00			
tax	NUMERIC(12,2)	NOT NULL	0.00			
quantity	SMALLINT	NOT NULL				

Column Name	Data Type	Nullable	Default	Key	References	Notes
subtotal	NUMERIC(12,2)	NOT NULL				
grand_total	NUMERIC(12,2)	NOT NULL				

Default ordering: id ASC · 21 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_orders_orderitem	id	Surrogate identity key.
UNIQUE	uq_orders_orderitem_uuid	uuid	Column-level uniqueness.
CHECK	orderitem_quantity_positive		((quantity >= ', 1))
FOREIGN KEY	fk_orders_orderitem_order_id	order_id	REFERENCES orders_order(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_orders_orderitem_product_id	product_id	REFERENCES products_product(id) ON DELETE SET NULL ON UPDATE CASCADE
FOREIGN KEY	fk_orders_orderitem_variant_id	variant_id	REFERENCES products_productvariant(id) ON DELETE SET NULL ON UPDATE CASCADE
INDEX	orderitem_order_idx	order	Composite B-tree.
INDEX	orderitem_sku_idx	sku	Composite B-tree.

Attribute	Detail
Purpose	One purchased line, fully snapshotted. product_name, brand_name, sku, size, colour, image_url and all money columns are frozen copies — the product FK is SET NULL precisely so a delisted product cannot erase what was sold.
Relationships	Child of orders_order (CASCADE). Optional references to Product and Variant (SET NULL). One-to-one target of reviews_review.order_item_id.
Used By APIs	Embedded in /orders/{order_number}; drives /reviews/pending
Used By Frontend Pages	Order Detail line items, Invoice PDF, Write a Review

orders_orderstatushistory [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; idx: statushistory_order_idx
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
order_id	BIGINT	NOT NULL		FK	orders_order	ON DELETE CASCADE; idx: statushistory_order_idx
previous_status	VARCHAR(20)	NOT NULL				pending confirmed processing packed shipped out_for_delivery delivered canc...
status	VARCHAR(20)	NOT NULL				pending confirmed processing packed shipped out_for_delivery delivered canc...
changed_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL; B-tree index
remarks	VARCHAR(255)	NOT NULL				

Default ordering: created_at ASC, id ASC · 9 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_orders_orderstatushistory	id	Surrogate identity key.
UNIQUE	uq_orders_orderstatushistory_uuid	uuid	Column-level uniqueness.
FOREIGN KEY	fk_orders_orderstatushistory_order_id	order_id	REFERENCES orders_order(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_orders_orderstatushistory_changed_by_id	changed_by_id	REFERENCES users_user(id) ON DELETE SET NULL ON UPDATE CASCADE

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
INDEX	statushistory_order_idx	order, created_at	Composite B-tree.

Attribute	Detail
Purpose	Append-only audit of every state transition. Rows are written once and never updated — a rewritable audit log answers nothing. This table, not the order's status column, is what the notification engine watches, which is what makes 'your order shipped' fire exactly once.
Relationships	Child of orders_order (CASCADE); optional actor reference to users_user (SET NULL).
Used By APIs	GET /orders/{order_number}/tracking
Used By Frontend Pages	Order Tracking timeline, Admin Order Detail

orders_shipment [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; idx: shipment_order_idx
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
order_id	BIGINT	NOT NULL		FK	orders_order	ON DELETE CASCADE; idx: shipment_order_idx
courier_name	VARCHAR(100)	NOT NULL				
tracking_number	VARCHAR(64)	NOT NULL				B-tree index
tracking_url	VARCHAR(500)	NOT NULL				
dispatched_at	TIMESTAMPTZ	NULL				
expected_delivery_date	DATE	NULL				
delivered_at	TIMESTAMPTZ	NULL				
remarks	VARCHAR(255)	NOT NULL				

Default ordering: created_at DESC · 12 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_orders_shipment	id	Surrogate identity key.
UNIQUE	uq_orders_shipment_uuid	uuid	Column-level uniqueness.
FOREIGN KEY	fk_orders_shipment_order_id	order_id	REFERENCES orders_order(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	shipment_order_idx	order, created_at DESC	Composite B-tree.

Attribute	Detail
Purpose	One dispatched parcel with courier and tracking reference. An order may have several — split shipments are normal in fashion retail.
Relationships	Child of orders_order (CASCADE).
Used By APIs	GET /orders/{order_number}/shipments · /orders/{order_number}/tracking
Used By Frontend Pages	Order Tracking, Admin Fulfilment

PROMOTIONS & COUPONS

coupons_coupon [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
code	VARCHAR(40)	NOT NULL		UQ		
description	VARCHAR(255)	NOT NULL				
discount_type	VARCHAR(16)	NOT NULL	percentage			flat percentage free_shipping
value	NUMERIC(12,2)	NOT NULL	0.00			
max_discount	NUMERIC(12,2)	NULL				
min_cart_value	NUMERIC(12,2)	NOT NULL	0.00			
currency	VARCHAR(3)	NOT NULL	INR			
max_uses	INTEGER	NOT NULL	0			
uses_per_user	SMALLINT	NOT NULL	1			
times_used	INTEGER	NOT NULL	0			
valid_from	TIMESTAMPTZ	NOT NULL	now()			idx: coupon_window_idx
valid_until	TIMESTAMPTZ	NULL				idx: coupon_window_idx
is_active	BOOLEAN	NOT NULL	TRUE			idx: coupon_window_idx, coupon_public_idx
is_public	BOOLEAN	NOT NULL	TRUE			idx: coupon_public_idx
first_order_only	BOOLEAN	NOT NULL	FALSE			

Default ordering: created_at DESC · 19 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_coupons_coupon	id	Surrogate identity key.
UNIQUE	uq_coupons_coupon_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_coupons_coupon_code	code	Column-level uniqueness.
CHECK	coupon_value_not_negative		((value >= ', Decimal('0')))
CHECK	coupon_min_cart_not_negative		((min_cart_value >= ', Decimal('0')))
CHECK	coupon_window_ordered		((valid_until IS NULL ', True), ('valid_until > ', F(valid_from)))
FOREIGN KEY	fk_coupons_coupon_categories	categories	REFERENCES catalog_category(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_coupons_coupon_brands	brands	REFERENCES catalog_brand(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_coupons_coupon_products	products	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	coupon_window_idx	is_active, valid_from, valid_until	Composite B-tree.
INDEX	coupon_public_idx	is_public, is_active	Composite B-tree.

Attribute	Detail
Purpose	Discount code definition: flat, percentage or free shipping, with a minimum cart value, an optional cap, global and per-customer usage limits and a validity window. A CHECK constraint guarantees valid_until is after valid_from, so an unusable window cannot be saved.

Relationships	Parent of coupons_couponusage. Optionally scoped to categories, brands or products through three bridge tables; an unscoped coupon applies store-wide.
Used By APIs	GET /coupons · /coupons/{code} · POST /coupons/validate · /coupons/apply · /coupons/remove · /cart/apply-coupon
Used By Frontend Pages	Cart → Apply Coupon, Checkout, Offers page, Admin Coupons

coupons_coupon_categories [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Surrogate key
coupon_id	BIGINT	NOT NULL		FK/UQ*	coupons_coupon	ON DELETE CASCADE; composite UQ with the other side
category_id	BIGINT	NOT NULL		FK/UQ*	catalog_category	ON DELETE CASCADE; composite UQ with the other side

Default ordering: None (unordered). · 3 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_coupons_coupon_categories	id	Surrogate identity key.
UNIQUE	uq_coupons_coupon_categories	coupon_id, category_id	One row per pair — the relationship itself carries no attributes.
FOREIGN KEY	fk_coupons_coupon_categories_coupon_id	coupon_id	REFERENCES coupons_coupon(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_coupons_coupon_categories_category_id	category_id	REFERENCES catalog_category(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	idx_coupons_coupon_categories_category_id	category_id	Reverse traversal; the composite UQ already covers the forward direction.

Attribute	Detail
Purpose	Bridge restricting a coupon to specific categories.
Relationships	Bridge: coupons_coupon ↔ catalog_category.
Used By APIs	Evaluated inside POST /coupons/validate
Used By Frontend Pages	Cart coupon validation

coupons_coupon_brands [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Surrogate key
coupon_id	BIGINT	NOT NULL		FK/UQ*	coupons_coupon	ON DELETE CASCADE; composite UQ with the other side
brand_id	BIGINT	NOT NULL		FK/UQ*	catalog_brand	ON DELETE CASCADE; composite UQ with the other side

Default ordering: None (unordered). · 3 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_coupons_coupon_brands	id	Surrogate identity key.
UNIQUE	uq_coupons_coupon_brands	coupon_id, brand_id	One row per pair — the relationship itself carries no attributes.
FOREIGN KEY	fk_coupons_coupon_brands_coupon_id	coupon_id	REFERENCES coupons_coupon(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_coupons_coupon_brands_brand_id	brand_id	REFERENCES catalog_brand(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	idx_coupons_coupon_brands_brand_id	brand_id	Reverse traversal; the composite UQ already covers the forward direction.

Attribute	Detail
Purpose	Bridge restricting a coupon to specific brands.
Relationships	Bridge: coupons_coupon ↔ catalog_brand.

Used By APIs	Evaluated inside POST /coupons/validate
Used By Frontend Pages	Cart coupon validation

coupons_coupon_products [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Surrogate key
coupon_id	BIGINT	NOT NULL		FK/UQ*	coupons_coupon	ON DELETE CASCADE; composite UQ with the other side
product_id	BIGINT	NOT NULL		FK/UQ*	products_product	ON DELETE CASCADE; composite UQ with the other side

Default ordering: None (unordered). · 3 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_coupons_coupon_products	id	Surrogate identity key.
UNIQUE	uq_coupons_coupon_products	coupon_id, product_id	One row per pair — the relationship itself carries no attributes.
FOREIGN KEY	fk_coupons_coupon_products_coupon_id	coupon_id	REFERENCES coupons_coupon(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_coupons_coupon_products_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	idx_coupons_coupon_products_product_id	product_id	Reverse traversal; the composite UQ already covers the forward direction.

Attribute	Detail
Purpose	Bridge restricting a coupon to specific products.
Relationships	Bridge: coupons_coupon ↔ products_product.
Used By APIs	Evaluated inside POST /coupons/validate
Used By Frontend Pages	Cart coupon validation

coupons_couponusage [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; idx: couponusage_user_idx
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
coupon_id	BIGINT	NOT NULL		FK	coupons_coupon	ON DELETE CASCADE; idx: couponusage_coupon_user_idx
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE; idx: couponusage_coupon_user_idx, couponusage_user_idx
order_id	BIGINT	NULL		FK	orders_order	ON DELETE CASCADE; B-tree index
discount_amount	NUMERIC(12, 2)	NOT NULL	0.00			
is_released	BOOLEAN	NOT NULL	FALSE			B-tree index
released_at	TIMESTAMPTZ	NULL				

Default ordering: created_at DESC · 10 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_coupons_couponusage	id	Surrogate identity key.
UNIQUE	uq_coupons_couponusage_uuid	uuid	Column-level uniqueness.

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
UNIQUE	unique_coupon_usage_per_order		COLUMNS ('coupon', 'order') (('order IS NULL ', False))
FOREIGN KEY	fk_coupons_couponusage_coupon_id	coupon_id	REFERENCES coupons_coupon(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_coupons_couponusage_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_coupons_couponusage_order_id	order_id	REFERENCES orders_order(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	couponusage_coupon_user_idx	coupon, user	Composite B-tree.
INDEX	couponusage_user_idx	user, created_at DESC	Composite B-tree.

Attribute	Detail
Purpose	Redemption ledger — the enforcement point for per-customer limits. A partial unique index on (coupon, order) stops a retried checkout from double-counting a redemption. is_released reverses a redemption when an order is cancelled.
Relationships	Child of Coupon and User (CASCADE) and optionally Order (CASCADE).
Used By APIs	GET /coupons/usages
Used By Frontend Pages	My Account → Coupons, Admin Coupon Report

PAYMENTS & REFUNDS

payments_payment [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; idx: payment_order_idx, payment_status_idx
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
order_id	BIGINT	NOT NULL		FK	orders_order	ON DELETE PROTECT; idx: payment_order_idx
gateway	VARCHAR(32)	NOT NULL				idx: payment_gateway_idx
method	VARCHAR(32)	NOT NULL				
gateway_order_id	VARCHAR(100)	NOT NULL				B-tree index
gateway_payment_id	VARCHAR(100)	NOT NULL				B-tree index
gateway_signature	VARCHAR(255)	NOT NULL				
transaction_id	VARCHAR(100)	NOT NULL				B-tree index
reference_number	VARCHAR(64)	NOT NULL				
amount	NUMERIC(12,2)	NOT NULL				
refunded_amount	NUMERIC(12,2)	NOT NULL	0.00			
currency	VARCHAR(3)	NOT NULL	INR			
status	VARCHAR(24)	NOT NULL	created			created authorised captured failed cancelled refunded partially_refunded; idx: payment_status_idx, payment_gateway_idx
failure_reason	VARCHAR(255)	NOT NULL				
captured_at	TIMESTAMP TZ	NULL				
raw_response	JSONB	NOT NULL	'{}'::jsonb			

Default ordering: created_at DESC · 19 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_payments_payment	id	Surrogate identity key.
UNIQUE	uq_payments_payment_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_gateway_payment_id		COLUMNS ('gateway', 'gateway_payment_id') (('gateway_payment_id > ', ''))
UNIQUE	unique_gateway_order_id		COLUMNS ('gateway', 'gateway_order_id') (('gateway_order_id > ', ''))
CHECK	payment_refund_within_amount		((('refunded_amount <= ' , F(amount))))
CHECK	payment_amount_positive		((('amount > ' , Decimal('0'))))
FOREIGN KEY	fk_payments_payment_order_id	order_id	REFERENCES orders_order(id) ON DELETE PROTECT ON UPDATE CASCADE
INDEX	payment_order_idx	order, created_at DESC	Composite B-tree.
INDEX	payment_status_idx	status, created_at DESC	Composite B-tree.
INDEX	payment_gateway_idx	gateway, status	Composite B-tree.

Attribute	Detail
Purpose	One payment intent against an order. Two partial unique indexes on (gateway, gateway_payment_id) and (gateway, gateway_order_id) make gateway identifiers idempotent, and a CHECK keeps refunded_amount within the captured amount so a payment can never be over-refunded.
Relationships	Child of orders_order (PROTECT — a paid order cannot be deleted). Parent of PaymentAttempt, Refund and PaymentWebhookLog.

Used By APIs	GET /payments · /payments/gateways · POST /payments/create · /payments/verify · /payments/webhook/{gateway}
Used By Frontend Pages	Checkout → Payment, Payment Status, Order Detail, Admin Payments

payments_paymentattempt [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
payment_id	BIGINT	NOT NULL		FK	payments_payment	ON DELETE CASCADE; idx: attempt_payment_idx
attempt_number	SMALLINT	NOT NULL	1			idx: attempt_payment_idx
status	VARCHAR(16)	NOT NULL	started			started succeeded failed abandoned; B-tree index
failure_reason	VARCHAR(255)	NOT NULL				
failure_code	VARCHAR(64)	NOT NULL				
retry_count	SMALLINT	NOT NULL	0			
gateway_response	JSONB	NOT NULL	'{}'::jsonb			

Default ordering: payment ASC, attempt_number ASC · 11 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_payments_paymentattempt	id	Surrogate identity key.
UNIQUE	uq_payments_paymentattempt_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_attempt_number_per_payment		COLUMNS ('payment', 'attempt_number')
FOREIGN KEY	fk_payments_paymentattempt_payment_id	payment_id	REFERENCES payments_payment(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	attempt_payment_idx	payment, attempt_number	Composite B-tree.

Attribute	Detail
Purpose	One try within a payment intent. A customer who fails on a card and then succeeds on UPI produces two attempts under one payment; without this table the failure would be invisible and the drop-off unmeasurable.
Relationships	Child of payments_payment (CASCADE). Unique on (payment, attempt_number).
Used By APIs	Embedded in payment detail responses.
Used By Frontend Pages	Admin Payment Detail

payments_refund [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; idx: refund_payment_idx, refund_status_idx
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
payment_id	BIGINT	NOT NULL		FK	payments_payment	ON DELETE PROTECT; idx: refund_payment_idx
amount	NUMERIC(12,2)	NOT NULL				
currency	VARCHAR(3)	NOT NULL	INR			

Column Name	Data Type	Nullable	Default	Key	References	Notes
status	VARCHAR(16)	NOT NULL	pending			pending processing processed failed; idx: refund_status_idx
reason	VARCHAR(24)	NOT NULL	other			order_cancelled order_returned item_unavailable damaged goodwill duplicate other; B-tree index
notes	VARCHAR(255)	NOT NULL				
gateway_refund_id	VARCHAR(100)	NOT NULL				B-tree index
reference_number	VARCHAR(64)	NOT NULL				
processed_at	TIMESTAMPTZ	NULL				
raw_response	JSONB	NOT NULL	'{}'::jsonb			

Default ordering: created_at DESC · 14 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_payments_refund	id	Surrogate identity key.
UNIQUE	uq_payments_refund_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_gateway_refund_id		COLUMNS ('gateway_refund_id') (('gateway_refund_id > ', ''))
CHECK	refund_amount_positive		((amount > ', Decimal('0')))
FOREIGN KEY	fk_payments_refund_payment_id	payment_id	REFERENCES payments_payment(id) ON DELETE PROTECT ON UPDATE CASCADE
INDEX	refund_payment_idx	payment, created_at DESC	Composite B-tree.
INDEX	refund_status_idx	status, created_at DESC	Composite B-tree.

Attribute	Detail
Purpose	Money returned, whole or partial. reason is stored because finance treats the categories differently: a cancellation is a sale that never happened, a return is a sale reversed, and goodwill is a marketing cost.
Relationships	Child of payments_payment (PROTECT).
Used By APIs	GET /refunds · POST /payments/refund
Used By Frontend Pages	Order Detail → Refund status, Admin Refund Queue

payments_paymentwebhooklog [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; idx: webhook_recent_idx
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
gateway	VARCHAR(32)	NOT NULL				idx: webhook_type_idx
event_id	VARCHAR(200)	NOT NULL				B-tree index
event_type	VARCHAR(64)	NOT NULL				idx: webhook_type_idx
signature	VARCHAR(255)	NOT NULL				
is_verified	BOOLEAN	NOT NULL	FALSE			
is_duplicate	BOOLEAN	NOT NULL	FALSE			B-tree index
payload	JSONB	NOT NULL	'{}'::jsonb			
headers	JSONB	NOT NULL	'{}'::jsonb			
processed_at	TIMESTAMPTZ	NULL				

Column Name	Data Type	Nullable	Default	Key	References	Notes
error	TEXT	NOT NULL				
payment_id	BIGINT	NULL		FK	payments_payment	ON DELETE SET NULL; B-tree index

Default ordering: created_at DESC · 15 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_payments_paymentwebhooklog	id	Surrogate identity key.
UNIQUE	uq_payments_paymentwebhooklog_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_webhook_event		COLUMNS ('gateway', 'event_id')
FOREIGN KEY	fk_payments_paymentwebhooklog_payment_id	payment_id	REFERENCES payments_payment(id) ON DELETE SET NULL ON UPDATE CASCADE
INDEX	webhook_type_idx	gateway, event_type	Composite B-tree.
INDEX	webhook_recent_idx	created_at DESC	Composite B-tree.

Attribute	Detail
Purpose	Every inbound gateway callback, verified or not. The unique constraint on (gateway, event_id) is the idempotency mechanism: a duplicate delivery raises a unique violation, which the application treats as 'already seen' rather than re-applying the event.
Relationships	Optional child of payments_payment (SET NULL) — an unmatched event is still logged.
Used By APIs	POST /payments/webhook/{gateway}
Used By Frontend Pages	Admin → Webhook Log

REVIEWS & RATINGS

reviews_review [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; idx: review_product_idx, review_user_idx
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; idx: review_product_idx, review_helpful_idx
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE; idx: review_user_idx
order_item_id	BIGINT	NULL		FK/UQ	orders_orderitem	ON DELETE SET NULL; one-to-one
rating	SMALLINT	NOT NULL				1 2 3 4 5; B-tree index
title	VARCHAR(120)	NOT NULL				
body	TEXT	NOT NULL				
is_verified_purchase	BOOLEAN	NOT NULL	FALSE			B-tree index
status	VARCHAR(12)	NOT NULL	pending			pending approved rejected; idx: review_product_idx, review_helpful_idx
moderated_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL; B-tree index
moderated_at	TIMESTAMP TZ	NULL				
rejection_reason	VARCHAR(255)	NOT NULL				
helpful_count	INTEGER	NOT NULL	0			idx: review_helpful_idx

Default ordering: created_at DESC · 16 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_reviews_review	id	Surrogate identity key.
UNIQUE	uq_reviews_review_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_reviews_review_order_item_id	order_item_id	Column-level uniqueness.
UNIQUE	unique_review_per_order_item		COLUMNS ('order_item') (('order_item IS NULL ', False))
CHECK	review_rating_within_range		((rating >= ' 1), (rating <= ' 5))
FOREIGN KEY	fk_reviews_review_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_reviews_review_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_reviews_review_order_item_id	order_item_id	REFERENCES orders_orderitem(id) ON DELETE SET NULL ON UPDATE CASCADE
FOREIGN KEY	fk_reviews_review_moderated_by_id	moderated_by_id	REFERENCES users_user(id) ON DELETE SET NULL ON UPDATE CASCADE
INDEX	review_product_idx	product, status, created_at DESC	Composite B-tree.
INDEX	review_helpful_idx	product, status, helpful_count DESC	Composite B-tree.
INDEX	review_user_idx	user, created_at DESC	Composite B-tree.
INDEX	review_queue_idx	status, created_at DESC	Composite B-tree.

Attribute	Detail
Purpose	A verified-purchase review. The unique constraint on order_item_id is the rule the module exists to protect: one review per purchased line. It is enforced in the database because a double-tapped submit is two concurrent requests and a check-then-insert loses that race. Only status = 'approved' rows feed the product rating.

Relationships	Child of Product and User (CASCADE), one-to-one with OrderItem (SET NULL), optional moderator reference to User (SET NULL). Parent of ReviewImage and HelpfulVote.
Used By APIs	GET /products/{slug}/reviews · /reviews/summary · /reviews/gallery · /reviews/eligibility · GET POST /reviews · PATCH DELETE /reviews/{uuid} · GET /reviews/pending · POST /reviews/{uuid}/helpful · GET /admin/reviews · POST /admin/reviews/moderate
Used By Frontend Pages	Product Detail → Reviews, Write a Review, My Reviews, Admin Moderation

reviews_reviewimage [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
review_id	BIGINT	NOT NULL		FK	reviews_review	ON DELETE CASCADE; idx: reviewimage_order_idx
image	VARCHAR(100)	NOT NULL				
thumbnail	VARCHAR(100)	NULL				
caption	VARCHAR(150)	NOT NULL				
display_order	SMALLINT	NOT NULL	0			idx: reviewimage_order_idx

Default ordering: display_order ASC, id ASC · 9 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_reviews_reviewimage	id	Surrogate identity key.
UNIQUE	uq_reviews_reviewimage_uuid	uuid	Column-level uniqueness.
FOREIGN KEY	fk_reviews_reviewimage_review_id	review_id	REFERENCES reviews_review(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	reviewimage_order_idx	review, display_order	Composite B-tree.

Attribute	Detail
Purpose	Customer photographs attached to a review, with a generated thumbnail.
Relationships	Child of reviews_review (CASCADE). Capped at five per review by the service layer.
Used By APIs	GET /products/{slug}/reviews/gallery
Used By Frontend Pages	Product Detail → Customer Photos, Review cards

reviews_helpfulvote [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
review_id	BIGINT	NOT NULL		FK	reviews_review	ON DELETE CASCADE; idx: helpfulvote_lookup_idx
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE; idx: helpfulvote_lookup_idx

Default ordering: created_at DESC · 6 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_reviews_helpfulvote	id	Surrogate identity key.
UNIQUE	uq_reviews_helpfulvote_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_helpful_vote_per_user		COLUMNS ('review', 'user')
FOREIGN KEY	fk_reviews_helpfulvote_review_id	review_id	REFERENCES reviews_review(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_reviews_helpfulvote_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	helpfulvote_lookup_idx	review, user	Composite B-tree.

Attribute	Detail
Purpose	One upvote by one customer on one review. Stored as rows rather than only a counter because the UI asks 'have I voted on this?', which an integer cannot answer. Review.helpful_count exists alongside purely so the list can be ordered without a join.
Relationships	Child of Review and User (CASCADE). Unique on (review, user).
Used By APIs	POST /reviews/{uuid}/helpful
Used By Frontend Pages	Product Detail → Was this helpful?

RECOMMENDATIONS & PERSONALISATION

recommendations_recentlyviewed [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
user_id	BIGINT	NULL		FK	users_user	ON DELETE CASCADE; idx: recentview_user_idx
session_key	VARCHAR(64)	NOT NULL				idx: recentview_session_idx
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; B-tree index
viewed_at	TIMESTAMPTZ	NOT NULL				idx: recentview_user_idx, recentview_session_idx
view_count	INTEGER	NOT NULL	1			

Default ordering: viewed_at DESC · 9 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_recommendations_recentlyviewed	id	Surrogate identity key.
UNIQUE	uq_recommendations_recentlyviewed_uuid	uuid	Column-level uniqueness.
CHECK	recentlyviewed_owned_by_user_xor_session		((('session_key', ''), ('user IS NULL ', False)), (('session_key > ', ''), ('user IS NULL ', True)))
UNIQUE	unique_recent_view_per_user		COLUMNS ('user', 'product') (('user IS NULL ', False))
UNIQUE	unique_recent_view_per_session		COLUMNS ('session_key', 'product') (('user IS NULL ', True))
FOREIGN KEY	fk_recommendations_recentlyviewed_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_recommendations_recentlyviewed_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	recentview_user_idx	user, viewed_at DESC	Composite B-tree.
INDEX	recentview_session_idx	session_key, viewed_at DESC	Composite B-tree.
INDEX	recentview_cleanup_idx	viewed_at	Composite B-tree.

Attribute	Detail
Purpose	Server-side browsing trail, owned by a user or a guest session under the same exclusive-or CHECK the cart uses. Capped at 30 rows per shopper by the application. Server-side rather than localStorage so the trail follows a shopper across devices and so 'customers also viewed' has anything to read.
Relationships	Optional child of User (CASCADE); child of Product (CASCADE).
Used By APIs	GET POST /recently-viewed · DELETE /recently-viewed/clear · /recently-viewed/item/{slug} · GET /products/recently-viewed
Used By Frontend Pages	Recently Viewed rail (Homepage, Product Detail, Listing)

recommendations_productaffinity [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		

Column Name	Data Type	Nullable	Default	Key	References	Notes
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; idx: affinity_lookup_idx
related_product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE; B-tree index
co_purchase_count	INTEGER	NOT NULL	0			B-tree index
score	DOUBLE PRECISION	NOT NULL	0.0			idx: affinity_lookup_idx
computed_at	TIMESTAMPTZ	NOT NULL				

Default ordering: score DESC, co_purchase_count DESC · 9 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_recommendations_productaffinity	id	Surrogate identity key.
UNIQUE	uq_recommendations_productaffinity_uuid	uuid	Column-level uniqueness.
UNIQUE	unique_product_affinity		COLUMNS ('product', 'related_product')
CHECK	affinity_not_self_referential		(NOT (('product', F(related_product))))
FOREIGN KEY	fk_recommendations_productaffinity_product_id	product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
FOREIGN KEY	fk_recommendations_productaffinity_related_product_id	related_product_id	REFERENCES products_product(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	affinity_lookup_idx	product, score DESC	Composite B-tree.

Attribute	Detail
Purpose	Materialised co-purchase graph behind 'frequently bought together'. Directed: A→B and B→A are separate rows, because a scarf sold with every coat is a strong suggestion on the coat and a weak one on the scarf. Rebuilt nightly by a Celery task; a self-referential edge is refused by CHECK.
Relationships	Two references to products_product (CASCADE). Unique on (product, related_product).
Used By APIs	GET /recommendations/products/{slug}/frequently-bought-together · POST /admin/recommendations/rebuild-affinities · GET /admin/recommendations/diagnose/{slug}
Used By Frontend Pages	Product Detail → Frequently Bought Together, Admin Diagnostics

NOTIFICATIONS

notifications_notification [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMPTZ	NOT NULL				auto-maintained; idx: notif_inbox_idx, notif_event_idx
updated_at	TIMESTAMPTZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE; idx: notif_inbox_idx, notif_dedupe_idx
event	VARCHAR(64)	NOT NULL				idx: notif_event_idx, notif_dedupe_idx
category	VARCHAR(16)	NOT NULL	system			account order payment review marketing system; B-tree index
channel	VARCHAR(8)	NOT NULL	in_app			in_app email sms push; idx: notif_queue_idx
subject	VARCHAR(200)	NOT NULL				
body	TEXT	NOT NULL				
link	VARCHAR(255)	NOT NULL				
context	JSONB	NOT NULL	'{}'::jsonb			
status	VARCHAR(12)	NOT NULL	pending			pending sent failed delivered; idx: notif_queue_idx
is_read	BOOLEAN	NOT NULL	FALSE			idx: notif_inbox_idx
read_at	TIMESTAMPTZ	NULL				
sent_at	TIMESTAMPTZ	NULL				
attempts	SMALLINT	NOT NULL	0			
error	VARCHAR(255)	NOT NULL				

Default ordering: created_at DESC · 18 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_notifications_notification	id	Surrogate identity key.
UNIQUE	uq_notifications_notification_uuid	uuid	Column-level uniqueness.
FOREIGN KEY	fk_notifications_notification_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE
INDEX	notif_inbox_idx	user, is_read, created_at DESC	Composite B-tree.
INDEX	notif_queue_idx	status, channel	Composite B-tree.
INDEX	notif_event_idx	event, created_at DESC	Composite B-tree.
INDEX	notif_dedupe_idx	user, event	Composite B-tree.

Attribute	Detail
Purpose	One message to one person on one channel. Simultaneously the in-app inbox row and the delivery ledger for email, SMS and push — one table rather than two so 'did we tell them?' has a single answer. The rendered context is kept as JSONB for support: what exactly did we send.
Relationships	Child of users_user (CASCADE).
Used By APIs	GET /notifications · /notifications/unread-count · POST /notifications/{uuid}/read · /notifications/read-all · DELETE /notifications/{uuid} · GET /admin/notifications · /admin/notifications/stats · POST /admin/notifications/drain · retry
Used By Frontend Pages	Notification bell menu, Notifications page, Admin Delivery Ledger

notifications_notificationpreference [IMPLEMENTED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
created_at	TIMESTAMP TZ	NOT NULL				auto-maintained; B-tree index
updated_at	TIMESTAMP TZ	NOT NULL				auto-maintained
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		
user_id	BIGINT	NOT NULL		FK/UQ	users_user	ON DELETE CASCADE; one-to-one
email_enabled	BOOLEAN	NOT NULL	TRUE			
sms_enabled	BOOLEAN	NOT NULL	TRUE			
push_enabled	BOOLEAN	NOT NULL	TRUE			
marketing_enabled	BOOLEAN	NOT NULL	TRUE			
push_token	VARCHAR(255)	NOT NULL				

Default ordering: None (unordered), · 10 columns

Type	Constraint / Index Name	Column(s)	Definition & Behaviour
PRIMARY KEY	pk_notifications_notificationpreference	id	Surrogate identity key.
UNIQUE	uq_notifications_notificationpreference_uuid	uuid	Column-level uniqueness.
UNIQUE	uq_notifications_notificationpreference_user_id	user_id	Column-level uniqueness.
FOREIGN KEY	fk_notifications_notificationpreference_user_id	user_id	REFERENCES users_user(id) ON DELETE CASCADE ON UPDATE CASCADE

Attribute	Detail
Purpose	Per-customer opt-outs, one row per account. Modelled as opt-out booleans rather than an opt-in subscription set so a newly added notification type reaches everyone by default. Marketing is the only category a customer may switch off wholesale; transactional mail is not a subscription.
Relationships	One-to-one with users_user (CASCADE).
Used By APIs	GET PATCH /notifications/preferences
Used By Frontend Pages	My Account → Notification Settings

LOOKUP & REFERENCE TABLES

lkp_color [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
hex_code	CHAR(7)	NOT NULL				#RRGGBB swatch
colour_family	VARCHAR(32)	NOT NULL				Blues, Neutrals — filter grouping
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

9 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Colour master. Today colour is a free-text VARCHAR(40) on products_productvariant with a separate hex column, which permits 'Navy', 'navy' and 'Navy Blue' to coexist and fragments the colour filter. Normalising it is the single highest-value lookup change in this document.
Relationships	Referenced by products_productvariant.color_id and the proposed variant_image table. Migration path: backfill from DISTINCT lower(trim(color)), then add the FK.
Used By APIs	GET /products/filters · embedded in /products/{slug}
Used By Frontend Pages	Listing colour filter, Product Detail swatches

lkp_size [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
size_system	VARCHAR(16)	NOT NULL	'ALPHA'			ALPHA UK EU US NUMERIC
sort_key	SMALLINT	NOT NULL	0			Canonical ordering across systems
category_id	BIGINT	NULL		FK	catalog_category	Scope: footwear vs apparel
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Size master with a sort key. Sizes are currently an enum on the variant, which cannot express that UK8 sits between XS and S, so size filters sort alphabetically and read as nonsense.
Relationships	Referenced by products_productvariant.size_id. Optional category scope so shoe sizes do not appear on a dress filter.
Used By APIs	GET /products/filters
Used By Frontend Pages	Listing size filter, Product Detail size picker, Size Guide

lkp_material [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
care_note	VARCHAR(255)	NOT NULL	' '			Default wash-care copy
is_sustainable	BOOLEAN	NOT NULL	FALSE			Drives the sustainability badge
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

9 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Fabric master, replacing the material enum on products_product.
Relationships	Referenced by products_product.material_id.
Used By APIs	GET /products/filters
Used By Frontend Pages	Listing fabric filter, Product Detail

lkp_occasion [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

7 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Occasion master (Casual, Formal, Wedding, Festive), replacing the product enum.
Relationships	Referenced by products_product.occasion_id and by lookbook.
Used By APIs	GET /products/filters
Used By Frontend Pages	Occasion filter, Shop by Occasion

lkp_gender [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

7 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Gender/audience master, shared by products and customer profiles.
Relationships	Referenced by products_product.gender_id and users_user.gender_id.
Used By APIs	GET /products/filters
Used By Frontend Pages	Gender nav, Profile form

Ikp_age_group [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
min_age_years	SMALLINT	NULL				Inclusive lower bound
max_age_years	SMALLINT	NULL				Inclusive upper bound
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

9 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Age banding (Adult, Teen, Kids 8–12, Infant). Absent from the current schema entirely — kidswear is only inferable from the category name.
Relationships	Referenced by products_product.age_group_id.
Used By APIs	GET /products/filters
Used By Frontend Pages	Kids landing, Age filter

Ikp_season [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name

Column Name	Data Type	Nullable	Default	Key	References	Notes
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

7 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Season master, replacing the season enum.
Relationships	Referenced by products_product.season_id and seasonal collections.
Used By APIs	GET /products/filters
Used By Frontend Pages	Seasonal campaign pages

Ikp_pattern [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

7 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Pattern master (Solid, Floral, Striped, Checked).
Relationships	Referenced by products_product.pattern_id.
Used By APIs	GET /products/filters
Used By Frontend Pages	Pattern filter

Ikp_fit [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

7 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Fit master (Slim, Regular, Relaxed, Oversized, Bodycon).
Relationships	Referenced by products_product.fit_id.
Used By APIs	GET /products/filters
Used By Frontend Pages	Fit filter, Product Detail

Ikp_country [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
iso2	CHAR(2)	NOT NULL		UQ		ISO 3166-1 alpha-2
iso3	CHAR(3)	NOT NULL		UQ		ISO 3166-1 alpha-3
name	VARCHAR(100)	NOT NULL				
phone_code	VARCHAR(8)	NOT NULL	' '			E.164 prefix
currency_code	CHAR(3)	NOT NULL	'INR'			ISO 4217
is_shipping_supported	BOOLEAN	NOT NULL	FALSE			Gates checkout
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	ISO country master for addresses, brand origin and product country of origin — all three are free-text VARCHAR(100) today.
Relationships	Referenced by users_address.country_id, catalog_brand.country_id and products_product.country_of_origin_id.
Used By APIs	GET /addresses (validation)
Used By Frontend Pages	Checkout country picker, Address form

INVENTORY & WAREHOUSING

warehouse [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
code	VARCHAR(24)	NOT NULL		UQ		e.g. BLR-01
name	VARCHAR(120)	NOT NULL				
address_line_1	VARCHAR(255)	NOT NULL				
address_line_2	VARCHAR(255)	NOT NULL	' '			
city	VARCHAR(100)	NOT NULL				
state	VARCHAR(100)	NOT NULL				
country_id	SMALLINT	NOT NULL		FK	lkp_country	ON DELETE RESTRICT
postal_code	VARCHAR(16)	NOT NULL				
latitude	NUMERIC(9,6)	NULL				Nearest-warehouse routing
longitude	NUMERIC(9,6)	NULL				
priority	SMALLINT	NOT NULL	0			Allocation preference; higher wins
is_pickup_enabled	BOOLEAN	NOT NULL	FALSE			Click-and-collect
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

17 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Physical fulfilment locations. The current schema holds a single stock integer per variant with no notion of where that stock sits, which makes multi-location fulfilment, transfers and location-aware delivery estimates impossible.
Relationships	Parent of inventory_stock, inventory_transaction and stock_alert.
Used By APIs	GET /admin/warehouses (new) · consumed by allocation during checkout
Used By Frontend Pages	Admin → Warehouses, Checkout delivery estimate

inventory_stock [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
variant_id	BIGINT	NOT NULL		FK	products_productvariant	ON DELETE CASCADE
warehouse_id	BIGINT	NOT NULL		FK	warehouse	ON DELETE RESTRICT
quantity_on_hand	INTEGER	NOT NULL	0			CHECK >= 0
quantity_reserved	INTEGER	NOT NULL	0			CHECK >= 0 AND <= on_hand
quantity_available	INTEGER	NOT NULL	GENERATED			GENERATED ALWAYS AS (quantity_on_hand - quantity_reserved) STORED
reorder_level	INTEGER	NOT NULL	5			Triggers stock_alert

Column Name	Data Type	Nullable	Default	Key	References	Notes
reorder_quantity	INTEGER	NOT NULL	0			Suggested purchase order size
bin_location	VARCHAR(32)	NOT NULL	' '			Aisle/shelf
last_counted_at	TIMESTAMPTZ	NULL				Last physical stocktake
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

12 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Stock on hand per variant per warehouse. This table takes over from products_productvariant.stock, which becomes a maintained roll-up. quantity_available is a generated column so no query can compute availability incorrectly.
Relationships	Composite UNIQUE (variant_id, warehouse_id). Roll-up of quantity_available across warehouses maintains products_productvariant.stock.
Used By APIs	GET /admin/analytics/inventory · /admin/dashboard/low-stock · /products/{slug}/availability
Used By Frontend Pages	Admin Inventory, Product Detail availability, Low Stock report

inventory_transaction [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
variant_id	BIGINT	NOT NULL		FK	products_productvariant	ON DELETE RESTRICT
warehouse_id	BIGINT	NOT NULL		FK	warehouse	ON DELETE RESTRICT
transaction_type	VARCHAR(24)	NOT NULL				purchase sale return reservation release adjustment transfer_in transfer_out damage
quantity_delta	INTEGER	NOT NULL				Signed; CHECK <> 0
balance_after	INTEGER	NOT NULL				Running balance for reconciliation
reference_type	VARCHAR(32)	NOT NULL	' '			order return manual po
reference_id	BIGINT	NULL				Polymorphic source id
order_item_id	BIGINT	NULL		FK	orders_orderitem	ON DELETE SET NULL
performed_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
notes	VARCHAR(255)	NOT NULL	' '			
created_at	TIMESTAMPTZ	NOT NULL	now()			Partition key

13 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Immutable ledger of every stock movement. The current schema mutates a counter with no history, so 'where did those forty units go?' is unanswerable. Append-only: corrections are new rows, never updates.
Relationships	Child of ProductVariant, Warehouse, OrderItem and User. Never updated or deleted.
Used By APIs	GET /admin/analytics/inventory · POST /admin/inventory/adjust (new)
Used By Frontend Pages	Admin Inventory Ledger, Stock Audit report

stock_alert [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
variant_id	BIGINT	NOT NULL		FK	products_productvariant	ON DELETE CASCADE
warehouse_id	BIGINT	NULL		FK	warehouse	ON DELETE CASCADE
alert_type	VARCHAR(24)	NOT NULL				low_stock out_of_stock back_in_stock_request
threshold	INTEGER	NULL				Level that fired the alert
quantity_at_alert	INTEGER	NULL				
subscriber_id	BIGINT	NULL		FK	users_user	ON DELETE CASCADE; back-in-stock only
is_resolved	BOOLEAN	NOT NULL	FALSE			
resolved_at	TIMESTAMPTZ	NULL				
notified_at	TIMESTAMPTZ	NULL				When the customer was told
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

12 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Open low-stock and out-of-stock signals, plus customer back-in-stock subscriptions. Today low stock is recomputed on every dashboard load with no record of whether anyone acted on it.
Relationships	Child of ProductVariant, Warehouse and User. Partial unique on (variant_id, subscriber_id) WHERE is_resolved = FALSE stops duplicate subscriptions.
Used By APIs	POST /products/{slug}/notify-me (new) · GET /admin/dashboard/low-stock
Used By Frontend Pages	Product Detail → Notify Me, Admin Low Stock queue

PRICING, OFFERS & PROMOTIONS

product_price_history [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE
variant_id	BIGINT	NULL		FK	products_productvariant	ON DELETE CASCADE
old_mrp	NUMERIC(12,2)	NULL				
new_mrp	NUMERIC(12,2)	NOT NULL				
old_selling_price	NUMERIC(12,2)	NULL				
new_selling_price	NUMERIC(12,2)	NOT NULL				
change_reason	VARCHAR(32)	NOT NULL	'manual'			manual flash_sale rule markdown
changed_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
effective_from	TIMESTAMPTZ	NOT NULL	now()			
created_at	TIMESTAMPTZ	NOT NULL	now()			Partition key

11 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Append-only price change log. Nothing in the current schema records what a product used to cost, so a 'was INR 2,999' claim cannot be substantiated — which is a legal exposure under Indian consumer-protection rules on reference pricing.
Relationships	Child of Product and Variant. Written by a trigger or the pricing service on any price update.
Used By APIs	GET /admin/products/{slug}/price-history (new)
Used By Frontend Pages	Admin Product Detail → Price History, Price-drop notifications

discount_rule [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
name	VARCHAR(150)	NOT NULL				
rule_type	VARCHAR(24)	NOT NULL				percentage flat bogo tiered bundle free_shipping
scope	VARCHAR(16)	NOT NULL	'product'			product category brand cart
value	NUMERIC(12,2)	NOT NULL	0.00			CHECK >= 0
max_discount	NUMERIC(12,2)	NULL				Cap for percentage rules
min_quantity	SMALLINT	NOT NULL	1			BOGO / tier trigger
min_cart_value	NUMERIC(12,2)	NOT NULL	0.00			
get_quantity	SMALLINT	NOT NULL	0			BOGO: units given free
priority	SMALLINT	NOT NULL	0			Resolution order when rules overlap
is_stackable	BOOLEAN	NOT NULL	FALSE			May combine with a coupon
valid_from	TIMESTAMPTZ	NOT NULL	now()			
valid_until	TIMESTAMPTZ	NULL				CHECK > valid_from

Column Name	Data Type	Nullable	Default	Key	References	Notes
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

17 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Automatic discounts that need no coupon code — 'Buy 2 Get 1', 'Flat 30% on Denim', tiered cart discounts. Currently expressible only as a per-product percentage, so every campaign is a bulk product edit that has to be undone by hand.
Relationships	Scoped through discount_rule_product / _category / _brand bridges (same shape as the coupon bridges). Evaluated by the cart pricing service before coupons.
Used By APIs	GET /offers (new) · evaluated inside GET /cart/summary
Used By Frontend Pages	Cart price breakdown, Offers page, Product Detail offer strip

offer [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
slug	VARCHAR(255)	NOT NULL		UQ		
title	VARCHAR(150)	NOT NULL				
subtitle	VARCHAR(255)	NOT NULL	''			
description	TEXT	NOT NULL	''			
banner_image	VARCHAR(255)	NULL				
badge_text	VARCHAR(32)	NOT NULL	''			e.g. 30% OFF
discount_rule_id	BIGINT	NULL		FK	discount_rule	ON DELETE SET NULL
coupon_id	BIGINT	NULL		FK	coupons_coupon	ON DELETE SET NULL
target_url	VARCHAR(500)	NOT NULL	''			
display_order	SMALLINT	NOT NULL	0			
is_active	BOOLEAN	NOT NULL	TRUE			
starts_at	TIMESTAMPTZ	NOT NULL	now()			
ends_at	TIMESTAMPTZ	NULL				
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

17 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Customer-facing promotional banner copy for a discount rule — the marketing surface, separate from the pricing mechanics so campaign copy can change without touching money logic.
Relationships	Optionally linked to a discount_rule or a coupon.
Used By APIs	GET /offers · /offers/{slug} (new)
Used By Frontend Pages	Offers page, Homepage offer strip, Product Detail offer box

flash_sale [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
slug	VARCHAR(255)	NOT NULL		UQ		
title	VARCHAR(150)	NOT NULL				
banner_image	VARCHAR(255)	NULL				
starts_at	TIMESTAMPTZ	NOT NULL				Drives the countdown timer
ends_at	TIMESTAMPTZ	NOT NULL				CHECK > starts_at
is_active	BOOLEAN	NOT NULL	TRUE			
total_stock_cap	INTEGER	NULL				Units across the whole event
units_sold	INTEGER	NOT NULL	0			Denormalised progress bar counter
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

12 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Time-boxed sale event with a countdown. The current /products/flash-sale endpoint just filters products discounted 40% or more — there is no event, no start and end time, and no per-item sale price or unit cap.
Relationships	Parent of flash_sale_item. EXCLUDE constraint on overlapping active windows prevents two live flash sales claiming the same product.
Used By APIs	GET /products/flash-sale · /flash-sales/{slug} (new)
Used By Frontend Pages	Homepage Flash Sale rail with countdown, Flash Sale landing

flash_sale_item [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
flash_sale_id	BIGINT	NOT NULL		FK	flash_sale	ON DELETE CASCADE
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE
variant_id	BIGINT	NULL		FK	products_productvariant	ON DELETE CASCADE
sale_price	NUMERIC(12, 2)	NOT NULL				CHECK > 0
unit_cap	INTEGER	NULL				Units released for this item
units_sold	INTEGER	NOT NULL	0			CHECK <= unit_cap
display_order	SMALLINT	NOT NULL	0			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	One product in a flash sale at an event-specific price and unit cap.
Relationships	Composite UNIQUE (flash_sale_id, product_id, variant_id).
Used By APIs	GET /products/flash-sale
Used By Frontend Pages	Flash Sale rail and landing page

SHIPPING, RETURNS & EXCHANGES

delivery_partner [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
code	VARCHAR(24)	NOT NULL		UQ		bluedart delhivery ekart
name	VARCHAR(120)	NOT NULL				
logo	VARCHAR(255)	NULL				
tracking_url_template	VARCHAR(500)	NOT NULL	' '			{tracking_number} placeholder
api_endpoint	VARCHAR(500)	NOT NULL	' '			Rate/manifest integration
supports_cod	BOOLEAN	NOT NULL	TRUE			Gates COD at checkout
supports_reverse_pickup	BOOLEAN	NOT NULL	TRUE			Required for returns
avg_delivery_days	SMALLINT	NOT NULL	5			Delivery estimate input
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

13 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Courier master. orders_shipment.courier_name is free text today, so the same courier appears under three spellings and per-courier performance cannot be measured.
Relationships	Referenced by orders_shipment.delivery_partner_id and return_request.pickup_partner_id.
Used By APIs	GET /orders/{order_number}/tracking · /checkout (delivery options)
Used By Frontend Pages	Order Tracking, Checkout delivery method, Admin Fulfilment

shipment_tracking_event [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
shipment_id	BIGINT	NOT NULL		FK	orders_shipment	ON DELETE CASCADE
status_code	VARCHAR(32)	NOT NULL				Courier's raw code
status_label	VARCHAR(120)	NOT NULL				Customer-facing text
location	VARCHAR(150)	NOT NULL	' '			Scan hub
remarks	VARCHAR(255)	NOT NULL	' '			
occurred_at	TIMESTAMPTZ	NOT NULL				Courier timestamp, not receipt time
raw_payload	JSONB	NOT NULL	'{}'::jsonb			Provider response, for disputes
created_at	TIMESTAMPTZ	NOT NULL	now()			

9 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Scan-by-scan courier events. orders_shipment currently holds only three timestamps, so the tracking page shows a coarse status where customers expect a scan history.

Relationships	Child of orders_shipment (CASCADE). UNIQUE (shipment_id, status_code, occurred_at) makes webhook replay idempotent.
Used By APIs	GET /orders/{order_number}/tracking · POST /webhooks/courier/{partner} (new)
Used By Frontend Pages	Order Tracking timeline

return_request [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
return_number	VARCHAR(32)	NOT NULL		UQ		Customer-facing RMA
order_id	BIGINT	NOT NULL		FK	orders_order	ON DELETE RESTRICT
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE RESTRICT
return_type	VARCHAR(16)	NOT NULL	'refund'			refund exchange store_credit
reason_code	VARCHAR(32)	NOT NULL				size_issue damaged wrong_item quality changed_mind late_delivery
reason_detail	VARCHAR(500)	NOT NULL	''			
status	VARCHAR(24)	NOT NULL	'requested'			requested approved rejected pickup_scheduled picked_up received qc_passed qc_failed completed
pickup_partner_id	BIGINT	NULL		FK	delivery_partner	ON DELETE SET NULL
pickup_tracking_number	VARCHAR(64)	NOT NULL	''			
pickup_scheduled_at	TIMESTAMPTZ	NULL				
received_at	TIMESTAMPTZ	NULL				
qc_notes	VARCHAR(500)	NOT NULL	''			Warehouse inspection outcome
refund_id	BIGINT	NULL		FK	payments_refund	ON DELETE SET NULL
refund_amount	NUMERIC(12, 2)	NOT NULL	0.00			CHECK >= 0
processed_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
completed_at	TIMESTAMPTZ	NULL				
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

20 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	A customer's request to send an item back. /orders/{order_number}/request-return exists today but only flips the order status — there is no return record, no per-item scope, no QC outcome and no link to the refund.
Relationships	Child of Order and User (RESTRICT). Parent of return_item. Optionally linked to the payments_refund it produced, closing the loop between goods and money.
Used By APIs	POST /orders/{order_number}/request-return · GET /returns · /admin/returns (new)
Used By Frontend Pages	Order Detail → Return, My Returns, Admin Returns queue

return_item [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
return_request_id	BIGINT	NOT NULL		FK	return_request	ON DELETE CASCADE
order_item_id	BIGINT	NOT NULL		FK	orders_orderitem	ON DELETE RESTRICT

Column Name	Data Type	Nullable	Default	Key	References	Notes
quantity	SMALLINT	NOT NULL	1			CHECK >= 1 AND <= ordered quantity
condition	VARCHAR(24)	NOT NULL	' '			unopened used damaged tags_removed
is_restockable	BOOLEAN	NOT NULL	FALSE			Set by QC; drives inventory_transaction
refund_amount	NUMERIC(12,2)	NOT NULL	0.00			Line share of the refund
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

9 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Which lines of an order are coming back, and how many units. Partial returns are the norm — a three-item order where one dress does not fit.
Relationships	Child of return_request (CASCADE) and orders_orderitem (RESTRICT). UNIQUE (return_request_id, order_item_id).
Used By APIs	Embedded in return endpoints
Used By Frontend Pages	Return form line selection, Admin QC screen

exchange_request [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
return_request_id	BIGINT	NOT NULL		FK	return_request	ON DELETE CASCADE
original_order_item_id	BIGINT	NOT NULL		FK	orders_orderitem	ON DELETE RESTRICT
requested_variant_id	BIGINT	NOT NULL		FK	products_productvariant	ON DELETE RESTRICT
quantity	SMALLINT	NOT NULL	1			CHECK >= 1
price_difference	NUMERIC(12,2)	NOT NULL	0.00			Signed; positive means customer pays
status	VARCHAR(24)	NOT NULL	'requested'			requested approved reserved dispatched completed rejected
replacement_order_id	BIGINT	NULL		FK	orders_order	ON DELETE SET NULL
reserved_until	TIMESTAMPTZ	NULL				Replacement stock hold expiry
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

12 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	A size or colour swap. Modelled separately from a refund because an exchange must reserve the replacement variant before the original comes back, or the customer returns an item for a size that has since sold out.
Relationships	Child of return_request. References the replacement variant and, once dispatched, the replacement order.
Used By APIs	POST /orders/{order_number}/request-exchange (new)
Used By Frontend Pages	Order Detail → Exchange, Admin Exchange queue

CONTENT, MERCHANDISING & CMS

homepage_banner [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
title	VARCHAR(150)	NOT NULL				
subtitle	VARCHAR(255)	NOT NULL	' '			
placement	VARCHAR(32)	NOT NULL	'hero'			hero strip grid_left grid_right footer category_top
desktop_image	VARCHAR(255)	NOT NULL				
mobile_image	VARCHAR(255)	NULL				Separate crop for small screens
alt_text	VARCHAR(200)	NOT NULL	' '			Accessibility; required for WCAG
cta_text	VARCHAR(40)	NOT NULL	' '			
target_url	VARCHAR(500)	NOT NULL	' '			
category_id	BIGINT	NULL		FK	catalog_category	ON DELETE SET NULL
display_order	SMALLINT	NOT NULL	0			
starts_at	TIMESTAMPTZ	NOT NULL	now()			
ends_at	TIMESTAMPTZ	NULL				
is_active	BOOLEAN	NOT NULL	TRUE			
click_count	INTEGER	NOT NULL	0			Denormalised CTR numerator
impression_count	INTEGER	NOT NULL	0			CTR denominator
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

19 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Scheduled promotional banners by placement. Homepage content is hard-coded in the frontend today, so every campaign needs a deploy.
Relationships	Optionally scoped to a category. Read by the homepage and category landing pages.
Used By APIs	GET /banners · /banners/{placement} (new)
Used By Frontend Pages	Homepage hero and strips, Category Landing banner

hero_carousel_slide [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
banner_id	BIGINT	NULL		FK	homepage_banner	ON DELETE CASCADE
title	VARCHAR(150)	NOT NULL				
subtitle	VARCHAR(255)	NOT NULL	' '			
desktop_image	VARCHAR(255)	NOT NULL				
mobile_image	VARCHAR(255)	NULL				

Column Name	Data Type	Nullable	Default	Key	References	Notes
video_url	VARCHAR(500)	NOT NULL	' '			Optional motion slide
cta_text	VARCHAR(40)	NOT NULL	' '			
target_url	VARCHAR(500)	NOT NULL	' '			
text_position	VARCHAR(16)	NOT NULL	'left'			left center right
duration_ms	INTEGER	NOT NULL	5000			Auto-advance delay
display_order	SMALLINT	NOT NULL	0			
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

16 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Ordered slides for the homepage hero carousel, with per-slide dwell time.
Relationships	Optionally derived from a homepage_banner.
Used By APIs	GET /homepage/carousel (new)
Used By Frontend Pages	Homepage hero carousel

product_video [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE
video_url	VARCHAR(500)	NOT NULL				CDN or provider URL
thumbnail	VARCHAR(255)	NULL				Poster frame
video_type	VARCHAR(16)	NOT NULL	'showcase'			showcase 360_spin try_on care
provider	VARCHAR(24)	NOT NULL	'self'			self youtube vimeo
duration_seconds	SMALLINT	NULL				
display_order	SMALLINT	NOT NULL	0			
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

12 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Product videos and 360° spins. products_productimage covers stills only, and video is now standard on fashion PDPs.
Relationships	Child of products_product (CASCADE).
Used By APIs	Embedded in GET /products/{slug}
Used By Frontend Pages	Product Detail gallery video tab

variant_image [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
variant_id	BIGINT	NOT NULL		FK	products_productvariant	ON DELETE CASCADE
image	VARCHAR(255)	NOT NULL				
thumbnail	VARCHAR(255)	NULL				
alt_text	VARCHAR(200)	NOT NULL	' '			
display_order	SMALLINT	NOT NULL	0			
is_primary	BOOLEAN	NOT NULL	FALSE			Partial UQ per variant
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Images tied to a specific colourway. products_productvariant has one image_override column, which cannot hold a gallery — so switching colour on the PDP cannot swap the whole image set.
Relationships	Child of products_productvariant (CASCADE). Partial unique on (variant_id) WHERE is_primary, mirroring products_productimage.
Used By APIs	Embedded in GET /products/{slug}
Used By Frontend Pages	Product Detail colour swatch → gallery swap

lookbook [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
slug	VARCHAR(255)	NOT NULL		UQ		
title	VARCHAR(150)	NOT NULL				
description	TEXT	NOT NULL	' '			
cover_image	VARCHAR(255)	NOT NULL				
season_id	SMALLINT	NULL		FK	lkp_season	ON DELETE SET NULL
occasion_id	SMALLINT	NULL		FK	lkp_occasion	ON DELETE SET NULL
influencer_id	BIGINT	NULL		FK	influencer	ON DELETE SET NULL
meta_title	VARCHAR(70)	NOT NULL	' '			SEO
meta_description	VARCHAR(170)	NOT NULL	' '			SEO
view_count	INTEGER	NOT NULL	0			
is_published	BOOLEAN	NOT NULL	FALSE			
published_at	TIMESTAMPTZ	NULL				
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

16 columns · NEW BUILD REQUIRED

Attribute	Detail
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Purpose	Editorial styling galleries — 'Wedding Season 2026', 'Office Capsule'. Pure merchandising content with no home in the current schema.
Relationships	Parent of lookbook_item. Optionally attributed to an influencer.
Used By APIs	GET /lookbooks · /lookbooks/{slug} (new)
Used By Frontend Pages	Lookbook index, Lookbook detail, Homepage editorial rail

lookbook_item [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
lookbook_id	BIGINT	NOT NULL		FK	lookbook	ON DELETE CASCADE
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE
image	VARCHAR(255)	NULL				Per-look shot
hotspot_x	NUMERIC(5,2)	NULL				0–100 % of width
hotspot_y	NUMERIC(5,2)	NULL				0–100 % of height
caption	VARCHAR(255)	NOT NULL	' '			
display_order	SMALLINT	NOT NULL	0			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	One shoppable hotspot on a lookbook image, positioned by percentage coordinates so it survives responsive scaling.
Relationships	Child of lookbook and Product. UNIQUE (lookbook_id, product_id, display_order).
Used By APIs	Embedded in GET /lookbooks/{slug}
Used By Frontend Pages	Lookbook detail shoppable hotspots

blog_category [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
description	TEXT	NOT NULL	' '			
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

8 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Taxonomy for the fashion blog.
Relationships	Parent of blog_post.

Used By APIs	GET /blog/categories (new)
Used By Frontend Pages	Blog index filter

blog_post [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
slug	VARCHAR(255)	NOT NULL		UQ		
title	VARCHAR(200)	NOT NULL				
excerpt	VARCHAR(500)	NOT NULL	' '			
content	TEXT	NOT NULL				Rich text or MDX
cover_image	VARCHAR(255)	NULL				
blog_category_id	SMALLINT	NULL		FK	blog_category	ON DELETE SET NULL
author_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
influencer_id	BIGINT	NULL		FK	influencer	ON DELETE SET NULL
reading_minutes	SMALLINT	NOT NULL	3			
view_count	INTEGER	NOT NULL	0			
meta_title	VARCHAR(70)	NOT NULL	' '			SEO
meta_description	VARCHAR(170)	NOT NULL	' '			SEO
is_published	BOOLEAN	NOT NULL	FALSE			
published_at	TIMESTAMPTZ	NULL				Index for the archive
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

18 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Fashion journal articles — SEO surface and top-of-funnel traffic.
Relationships	Child of blog_category, User and Influencer. Bridged to products via blog_post_products for shoppable articles.
Used By APIs	GET /blog · /blog/{slug} (new)
Used By Frontend Pages	Blog index, Blog article, Homepage editorial rail

influencer [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
slug	VARCHAR(255)	NOT NULL		UQ		
name	VARCHAR(150)	NOT NULL				
handle	VARCHAR(80)	NOT NULL		UQ		Social handle
bio	TEXT	NOT NULL	' '			
avatar	VARCHAR(255)	NULL				

Column Name	Data Type	Nullable	Default	Key	References	Notes
user_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL; if they have a login
instagram_url	VARCHAR(500)	NOT NULL	' '			
youtube_url	VARCHAR(500)	NOT NULL	' '			
follower_count	INTEGER	NOT NULL	0			
commission_percentage	NUMERIC(5,2)	NOT NULL	0.00			Affiliate rate
referral_code	VARCHAR(24)	NOT NULL	' '	UQ		Attribution code
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

16 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Creator and brand-ambassador profiles behind lookbooks, articles and affiliate attribution.
Relationships	Parent of lookbook, blog_post and influencer_campaign.
Used By APIs	GET /influencers · /influencers/{slug} (new)
Used By Frontend Pages	Influencer landing, Lookbook credits, Blog byline

influencer_campaign [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
influencer_id	BIGINT	NOT NULL		FK	influencer	ON DELETE CASCADE
name	VARCHAR(150)	NOT NULL				
coupon_id	BIGINT	NULL		FK	coupons_coupon	ON DELETE SET NULL
landing_url	VARCHAR(500)	NOT NULL	' '			
budget	NUMERIC(12,2)	NOT NULL	0.00			
clicks	INTEGER	NOT NULL	0			Attribution counters
orders_count	INTEGER	NOT NULL	0			
revenue_attributed	NUMERIC(14,2)	NOT NULL	0.00			
commission_earned	NUMERIC(12,2)	NOT NULL	0.00			
starts_at	TIMESTAMPTZ	NOT NULL	now()			
ends_at	TIMESTAMPTZ	NULL				
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

16 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	A paid or affiliate collaboration with measurable attribution.
Relationships	Child of influencer; optionally bound to a coupon for attribution.

Used By APIs	GET /admin/influencer-campaigns (new)
Used By Frontend Pages	Admin Marketing → Campaigns

cms_page [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
slug	VARCHAR(255)	NOT NULL		UQ		
title	VARCHAR(200)	NOT NULL				
content	TEXT	NOT NULL				Rich text
page_type	VARCHAR(24)	NOT NULL	'static'			static legal help landing
meta_title	VARCHAR(70)	NOT NULL	''			SEO
meta_description	VARCHAR(170)	NOT NULL	''			SEO
version	INTEGER	NOT NULL	1			Incremented on publish
effective_from	TIMESTAMPTZ	NULL				Legal pages need dated versions
is_published	BOOLEAN	NOT NULL	FALSE			
published_at	TIMESTAMPTZ	NULL				
updated_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

15 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Editable static pages — About, Privacy, Terms, Shipping Policy, Size Guide. Currently frontend files, so a legal copy change is a deploy.
Relationships	Standalone. Legal pages should retain superseded versions for audit.
Used By APIs	GET /pages/{slug} (new)
Used By Frontend Pages	About, Privacy Policy, Terms, Shipping Policy, Size Guide

faq_category [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		Stable machine key
label	VARCHAR(80)	NOT NULL				Display name
display_order	SMALLINT	NOT NULL	0			Sort position in filter rails
is_active	BOOLEAN	NOT NULL	TRUE			Soft disable
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

7 columns · NEW BUILD REQUIRED

Attribute	Detail
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Purpose	Grouping for FAQs (Orders, Returns, Payments, Sizing).
Relationships	Parent of faq.
Used By APIs	GET /faqs (new)
Used By Frontend Pages	Help Centre navigation

faq [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
faq_category_id	SMALLINT	NULL		FK	faq_category	ON DELETE SET NULL
question	VARCHAR(300)	NOT NULL				
answer	TEXT	NOT NULL				
display_order	SMALLINT	NOT NULL	0			
helpful_count	INTEGER	NOT NULL	0			
not_helpful_count	INTEGER	NOT NULL	0			
is_published	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

11 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Question and answer pairs for the help centre, with a helpfulness signal.
Relationships	Child of faq_category.
Used By APIs	GET /faqs · /faqs/{category} (new)
Used By Frontend Pages	Help Centre, Product Detail FAQ

product_question [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE
question	VARCHAR(500)	NOT NULL				
status	VARCHAR(12)	NOT NULL	'pending'			pending approved rejected
answer_count	SMALLINT	NOT NULL	0			Denormalised
upvote_count	INTEGER	NOT NULL	0			Sorts the most-asked to the top
moderated_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
moderated_at	TIMESTAMPTZ	NULL				
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

12 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Customer questions on a product page. Distinct from reviews: a question needs no purchase and expects an answer.
Relationships	Child of Product and User. Parent of product_answer. Moderated on the same queue pattern as reviews_review.
Used By APIs	GET POST /products/{slug}/questions (new)
Used By Frontend Pages	Product Detail → Questions & Answers

product_answer [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
question_id	BIGINT	NOT NULL		FK	product_question	ON DELETE CASCADE
user_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
answer	VARCHAR(1000)	NOT NULL				
answered_by_type	VARCHAR(16)	NOT NULL	'customer'			customer seller support
is_verified_buyer	BOOLEAN	NOT NULL	FALSE			Derived from order history
is_official	BOOLEAN	NOT NULL	FALSE			Pinned above customer answers
helpful_count	INTEGER	NOT NULL	0			
status	VARCHAR(12)	NOT NULL	'pending'			pending approved rejected
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

12 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	An answer from a verified buyer, a seller or support. answered_by_type is what lets the UI badge an official reply differently from a shopper's opinion.
Relationships	Child of product_question (CASCADE) and User (SET NULL).
Used By APIs	POST /products/{slug}/questions/{id}/answers (new)
Used By Frontend Pages	Product Detail → Q&A thread

CUSTOMER ENGAGEMENT & LOYALTY

loyalty_account [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
user_id	BIGINT	NOT NULL		FK/UQ	users_user	ON DELETE CASCADE; one per customer
points_balance	INTEGER	NOT NULL	0			CHECK >= 0
lifetime_points_earned	INTEGER	NOT NULL	0			Drives tier
lifetime_points_redeemed	INTEGER	NOT NULL	0			
tier	VARCHAR(16)	NOT NULL	'bronze'			bronze silver gold platinum
tier_expires_at	TIMESTAMPTZ	NULL				Tiers lapse without activity
total_spent	NUMERIC(14, 2)	NOT NULL	0.00			Tier qualification input
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

11 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	One points balance per customer with a tier. Balance is a maintained roll-up of loyalty_transaction — the ledger is authoritative, this is the fast read.
Relationships	One-to-one with users_user. Parent of loyalty_transaction.
Used By APIs	GET /loyalty · /loyalty/balance (new)
Used By Frontend Pages	My Account → Rewards, Checkout points redemption

loyalty_transaction [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
loyalty_account_id	BIGINT	NOT NULL		FK	loyalty_account	ON DELETE CASCADE
transaction_type	VARCHAR(24)	NOT NULL				earn redeem expire adjust refund_reversal referral_bonus
points	INTEGER	NOT NULL				Signed; CHECK <> 0
balance_after	INTEGER	NOT NULL				Reconciliation aid
order_id	BIGINT	NULL		FK	orders_order	ON DELETE SET NULL
description	VARCHAR(255)	NOT NULL	' '			
expires_at	TIMESTAMPTZ	NULL				Earning rows only
created_at	TIMESTAMPTZ	NOT NULL	now()			Partition key

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Append-only points ledger. Points expire, so each earning row carries its own expiry and redemption consumes oldest-first.
Relationships	Child of loyalty_account and optionally Order. Never updated.

Used By APIs	GET /loyalty/transactions (new)
Used By Frontend Pages	My Account → Points History

gift_card [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
code_hash	VARCHAR(128)	NOT NULL		UQ		SHA-256; never store plaintext
code_last4	CHAR(4)	NOT NULL				Display only
initial_value	NUMERIC(12,2)	NOT NULL				CHECK > 0
current_balance	NUMERIC(12,2)	NOT NULL				CHECK >= 0 AND <= initial_value
currency	CHAR(3)	NOT NULL	'INR'			
purchased_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
recipient_email	VARCHAR(254)	NOT NULL	''			
recipient_name	VARCHAR(150)	NOT NULL	''			
message	VARCHAR(500)	NOT NULL	''			Gift note
redeemed_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
status	VARCHAR(16)	NOT NULL	'active'			active redeemed expired cancelled
issued_at	TIMESTAMPTZ	NOT NULL	now()			
expires_at	TIMESTAMPTZ	NULL				
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

17 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Stored-value cards. The code is stored only as a hash — a gift card code is a bearer instrument, and a leaked database dump of plaintext codes is a direct financial loss.
Relationships	References the purchaser and the redeemer. Parent of gift_card_transaction.
Used By APIs	POST /gift-cards/purchase · /gift-cards/redeem · GET /gift-cards/balance (new)
Used By Frontend Pages	Gift Card purchase, Checkout redemption, My Account → Gift Cards

gift_card_transaction [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
gift_card_id	BIGINT	NOT NULL		FK	gift_card	ON DELETE RESTRICT
order_id	BIGINT	NULL		FK	orders_order	ON DELETE SET NULL
transaction_type	VARCHAR(16)	NOT NULL				issue redeem refund expire
amount	NUMERIC(12,2)	NOT NULL				Signed
balance_after	NUMERIC(12,2)	NOT NULL				
created_at	TIMESTAMPTZ	NOT NULL	now()			

7 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Every debit and credit against a card balance.
Relationships	Child of gift_card (RESTRICT — a card with movements is never deleted).
Used By APIs	Embedded in gift card endpoints
Used By Frontend Pages	My Account → Gift Card history

referral [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
referrer_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE
referred_user_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
referral_code	VARCHAR(24)	NOT NULL				Referrer's code
referred_email	VARCHAR(254)	NOT NULL	' '			Before signup
status	VARCHAR(16)	NOT NULL	'invited'			invited signed_up qualified rewarded expired
qualifying_order_id	BIGINT	NULL		FK	orders_order	ON DELETE SET NULL
referrer_reward_points	INTEGER	NOT NULL	0			
referred_reward_points	INTEGER	NOT NULL	0			
rewarded_at	TIMESTAMPTZ	NULL				
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

13 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Refer-a-friend tracking. A partial unique index on referred_user_id stops one person being claimed by two referrers, which is the standard abuse.
Relationships	Two references to users_user. Reward payout writes to loyalty_transaction.
Used By APIs	GET /referrals · POST /referrals/invite (new)
Used By Frontend Pages	My Account → Refer a Friend, Signup with referral code

SEARCH, BEHAVIOUR & AI PERSONALISATION

search_history [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
user_id	BIGINT	NULL		FK	users_user	ON DELETE CASCADE
session_key	VARCHAR(64)	NOT NULL	' '			Guest identity
query	VARCHAR(255)	NOT NULL				Normalised lower-case
normalised_query	VARCHAR(255)	NOT NULL				Stemmed; groups variants
result_count	INTEGER	NOT NULL	0			0 = catalogue gap
clicked_product_id	BIGINT	NULL		FK	products_product	ON DELETE SET NULL
click_position	SMALLINT	NULL				Relevance signal
filters_applied	JSONB	NOT NULL	'{}'::jsonb			Facet state
device_type	VARCHAR(16)	NOT NULL	' '			mobile desktop tablet
created_at	TIMESTAMPTZ	NOT NULL	now()			Partition key

11 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Every search query with its result count. Nothing persists searches today, so the zero-result queries that reveal catalogue gaps are invisible.
Relationships	Optional child of User and Product. Feeds trending_search and the merchandising zero-results report.
Used By APIs	GET /products/search (writes) · GET /products/suggestions
Used By Frontend Pages	Search bar, Search Results, My Account → Recent Searches

trending_search [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
query	VARCHAR(255)	NOT NULL				
search_count	INTEGER	NOT NULL	0			
click_through_rate	NUMERIC(5,2)	NOT NULL	0.00			Quality signal
window_start	DATE	NOT NULL				
window_end	DATE	NOT NULL				
rank	SMALLINT	NOT NULL	0			Position in the window
is_promoted	BOOLEAN	NOT NULL	FALSE			Merchandiser pin
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Aggregated popular queries per window. /products/suggestions serves a static list today; this makes it reflect what people actually search.
Relationships	Derived from search_history by a scheduled job. UNIQUE (query, window_start, window_end).

Used By APIs	GET /products/suggestions · /products/search-defaults
Used By Frontend Pages	Search bar suggestions, Trending Searches chips

clickstream_event [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		Composite PK with occurred_at when partitioned
user_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
session_key	VARCHAR(64)	NOT NULL	''			
event_type	VARCHAR(32)	NOT NULL				page_view product_view add_to_cart remove_from_cart wishlist_add checkout_start purchase search filter banner_click
product_id	BIGINT	NULL		FK	products_product	ON DELETE SET NULL
category_id	BIGINT	NULL		FK	catalog_category	ON DELETE SET NULL
page_url	VARCHAR(500)	NOT NULL	''			
referrer_url	VARCHAR(500)	NOT NULL	''			
utm_source	VARCHAR(80)	NOT NULL	''			Attribution
utm_medium	VARCHAR(80)	NOT NULL	''			
utm_campaign	VARCHAR(120)	NOT NULL	''			
device_type	VARCHAR(16)	NOT NULL	''			
ip_address	INET	NULL				Truncate or hash under DPDP
metadata	JSONB	NOT NULL	'{}'::jsonb			Event-specific payload
occurred_at	TIMESTAMPTZ	NOT NULL	now()			PARTITION KEY

15 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Raw interaction events — the highest-volume table in the schema and the reason the partitioning section exists. Expect tens of millions of rows a month; it must be range-partitioned by day from the outset.
Relationships	Optional references to User, Product and Category — all SET NULL so a deletion never blocks on the largest table in the database.
Used By APIs	POST /events (new, fire-and-forget)
Used By Frontend Pages	Every page (instrumentation layer)

fashion_preference [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
user_id	BIGINT	NOT NULL		FK/UQ	users_user	ON DELETE CASCADE; one per customer
preferred_sizes	JSONB	NOT NULL	'{}'::jsonb			Per category: {"tops":"M"}
preferred_colours	JSONB	NOT NULL	'[]'::jsonb			Weighted colour ids
preferred_brands	JSONB	NOT NULL	'[]'::jsonb			Weighted brand ids
preferred_categories	JSONB	NOT NULL	'[]'::jsonb			Weighted category ids
preferred_occasions	JSONB	NOT NULL	'[]'::jsonb			
price_band_min	NUMERIC(12,2)	NULL				Observed spend floor

Column Name	Data Type	Nullable	Default	Key	References	Notes
price_band_max	NUMERIC(12,2)	NULL				Observed spend ceiling
style_vector	JSONB	NOT NULL	'{}'::jsonb			Embedding; migrate to pgvector at scale
is_explicit	BOOLEAN	NOT NULL	FALSE			Declared by the customer vs inferred
last_computed_at	TIMESTAMPTZ	NULL				Recompute staleness
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

15 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	A shopper's derived and declared style profile — the feature vector the recommendation engine reads. Today taste is recomputed from orders, wishlist and browsing on every request.
Relationships	One-to-one with users_user. Written by the nightly personalisation job; read by the for-you and new-for-you rails.
Used By APIs	GET /recommendations/for-you · /recommendations/new-for-you
Used By Frontend Pages	Homepage For You rail, Style Quiz, My Account → Preferences

recommendation_history [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
user_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
session_key	VARCHAR(64)	NOT NULL	' '			
product_id	BIGINT	NOT NULL		FK	products_product	ON DELETE CASCADE
source_product_id	BIGINT	NULL		FK	products_product	ON DELETE SET NULL
rail	VARCHAR(40)	NOT NULL				related similar fbt trending recently_popular also_viewed new_for_you for_you
algorithm_version	VARCHAR(24)	NOT NULL	'v1'			A/B comparison
score	DOUBLE PRECISION	NOT NULL	0			Model score at serve time
position	SMALLINT	NOT NULL	0			Slot in the rail
was_clicked	BOOLEAN	NOT NULL	FALSE			
was_added_to_cart	BOOLEAN	NOT NULL	FALSE			
was_purchased	BOOLEAN	NOT NULL	FALSE			
served_at	TIMESTAMPTZ	NOT NULL	now()			Partition key

13 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	What was recommended, where, and whether it was clicked or bought. Without this there is no way to tell whether the recommendation engine earns its place.
Relationships	References User and Product twice (recommended and source).
Used By APIs	Written by every /recommendations/* endpoint
Used By Frontend Pages	All recommendation rails (instrumentation)

CONTENT, MERCHANDISING & CMS

notification_template [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
event_key	VARCHAR(64)	NOT NULL		UQ*		Composite UQ with channel and locale
channel	VARCHAR(8)	NOT NULL		UQ*		in_app email sms push
locale	VARCHAR(10)	NOT NULL	'en'	UQ*		Localisation
category	VARCHAR(16)	NOT NULL	'system'			account order payment review marketing system
subject_template	VARCHAR(200)	NOT NULL				str.format placeholders
body_template	TEXT	NOT NULL				
link_template	VARCHAR(255)	NOT NULL	''			
available_variables	JSONB	NOT NULL	'[]'::jsonb			Documents valid placeholders
is_active	BOOLEAN	NOT NULL	TRUE			
version	INTEGER	NOT NULL	1			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

14 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Database-backed notification copy. Nineteen event templates live in Python today (apps/notifications/templates.py), which means marketing cannot change a subject line without a deploy. Moving them here keeps the code as the fallback.
Relationships	Read by the notification service before falling back to the code catalogue. Composite UNIQUE (event_key, channel, locale).
Used By APIs	GET /admin/notification-templates (new)
Used By Frontend Pages	Admin → Notification Templates

email_template [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
code	VARCHAR(64)	NOT NULL		UQ		
name	VARCHAR(150)	NOT NULL				
subject	VARCHAR(200)	NOT NULL				
html_body	TEXT	NOT NULL				Inlined CSS
text_body	TEXT	NOT NULL	''			Multipart alternative
layout_code	VARCHAR(64)	NOT NULL	''			Shared header/footer wrapper
from_name	VARCHAR(120)	NOT NULL	''			
reply_to	VARCHAR(254)	NOT NULL	''			
locale	VARCHAR(10)	NOT NULL	'en'			

Column Name	Data Type	Nullable	Default	Key	References	Notes
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMP TZ	NOT NULL	now()			UTC
updated_at	TIMESTAMP TZ	NOT NULL	now()			UTC; touched on write

14 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	HTML email layouts, separate from notification_template because an email has a wrapper, inline CSS and a plain-text alternative that a push payload does not.
Relationships	Referenced by notification_template for email-channel rows.
Used By APIs	GET /admin/email-templates (new)
Used By Frontend Pages	Admin → Email Templates

ADMINISTRATION, RBAC & AUDIT

admin_role [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(32)	NOT NULL		UQ		super_admin catalog_manager order_manager finance support marketing
name	VARCHAR(80)	NOT NULL				
description	VARCHAR(255)	NOT NULL	' '			
is_system	BOOLEAN	NOT NULL	FALSE			System roles cannot be deleted
is_active	BOOLEAN	NOT NULL	TRUE			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

8 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Named back-office roles. Authorisation is a single is_staff boolean today, so a catalogue merchandiser and a finance controller have identical access — including to revenue and refunds.
Relationships	Bridged to admin_permission and users_user.
Used By APIs	GET /admin/roles (new)
Used By Frontend Pages	Admin → Roles & Permissions

admin_permission [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	SMALLSERIAL	NOT NULL	identity	PK		
code	VARCHAR(64)	NOT NULL		UQ		e.g. orders.refund
module	VARCHAR(32)	NOT NULL				catalog orders payments reviews analytics cms
action	VARCHAR(24)	NOT NULL				view create update delete approve export
description	VARCHAR(255)	NOT NULL	' '			
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

7 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Atomic permission keys, one per module and action.
Relationships	Bridged to admin_role.
Used By APIs	GET /admin/permissions (new)
Used By Frontend Pages	Admin → Roles & Permissions

admin_role_permission [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
role_id	SMALLINT	NOT NULL		FK	admin_role	ON DELETE CASCADE
permission_id	SMALLINT	NOT NULL		FK	admin_permission	ON DELETE CASCADE
granted_at	TIMESTAMPTZ	NOT NULL	now()			
granted_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL

5 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Bridge granting permissions to a role.
Relationships	Bridge: admin_role ↔ admin_permission. UNIQUE (role_id, permission_id).
Used By APIs	Evaluated on every /admin/* request
Used By Frontend Pages	Admin → Role editor

admin_user_role [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE
role_id	SMALLINT	NOT NULL		FK	admin_role	ON DELETE CASCADE
assigned_at	TIMESTAMPTZ	NOT NULL	now()			
assigned_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
expires_at	TIMESTAMPTZ	NULL				Temporary elevation

6 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Bridge assigning back-office roles to staff accounts, with optional expiry for temporary elevation.
Relationships	Bridge: users_user ↔ admin_role. UNIQUE (user_id, role_id).
Used By APIs	Evaluated on every /admin/* request
Used By Frontend Pages	Admin → Staff

audit_log [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
entity_type	VARCHAR(64)	NOT NULL				Table name
entity_id	VARCHAR(64)	NOT NULL				Text, so UUID and BIGINT both fit
action	VARCHAR(24)	NOT NULL				create update delete approve reject export
old_values	JSONB	NULL				Changed columns only
new_values	JSONB	NULL				
changed_fields	JSONB	NOT NULL	'[]'::jsonb			Fast filter without diffing

Column Name	Data Type	Nullable	Default	Key	References	Notes
performed_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
performed_by_type	VARCHAR(16)	NOT NULL	'user'			user system api job
ip_address	INET	NULL				
user_agent	VARCHAR(500)	NOT NULL	' '			
request_id	VARCHAR(64)	NOT NULL	' '			Correlates with application logs
created_at	TIMESTAMPTZ	NOT NULL	now()			PARTITION KEY

14 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Row-level before/after change log for sensitive entities. There is currently no record of who changed a price, approved a refund or edited an order — a gap that fails any commercial audit.
Relationships	Optional reference to the acting user (SET NULL — the log outlives the account). Written by database triggers on sensitive tables, not by application code.
Used By APIs	GET /admin/audit-logs (new)
Used By Frontend Pages	Admin → Audit Trail

activity_log [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
user_id	BIGINT	NULL		FK	users_user	ON DELETE CASCADE
activity_type	VARCHAR(32)	NOT NULL				login logout login_failed password_change profile_update address_add order_place review_post
description	VARCHAR(255)	NOT NULL	' '			
ip_address	INET	NULL				
user_agent	VARCHAR(500)	NOT NULL	' '			
device_type	VARCHAR(16)	NOT NULL	' '			
location	VARCHAR(120)	NOT NULL	' '			Geo-IP city; 'new device' alerts
is_suspicious	BOOLEAN	NOT NULL	FALSE			Flagged by risk rules
created_at	TIMESTAMPTZ	NOT NULL	now()			PARTITION KEY

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Customer-facing activity feed: logins, password changes, address edits, order placement. Distinct from audit_log, which is for staff actions on data.
Relationships	Child of users_user (CASCADE).
Used By APIs	GET /account/activity (new)
Used By Frontend Pages	My Account → Login Activity, Security alerts

system_setting [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
key	VARCHAR(100)	NOT NULL		PK		Natural key
value	TEXT	NOT NULL				
data_type	VARCHAR(16)	NOT NULL	'string'			string integer decimal boolean json

Column Name	Data Type	Nullable	Default	Key	References	Notes
module	VARCHAR(32)	NOT NULL	''			cart catalog payments notifications
description	VARCHAR(255)	NOT NULL	''			
is_public	BOOLEAN	NOT NULL	FALSE			Safe to expose to the frontend
is_encrypted	BOOLEAN	NOT NULL	FALSE			Secrets stay in the environment
updated_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

10 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Runtime configuration a business user may change without a deploy. Everything is environment variables today, so altering the free-shipping threshold restarts the application.
Relationships	Standalone. Cached aggressively; invalidated on write.
Used By APIs	GET /admin/settings · GET /config/public (new)
Used By Frontend Pages	Admin → Settings, frontend bootstrap config

SUPPORT & COMMUNICATION

contact_message [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
user_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
name	VARCHAR(150)	NOT NULL				
email	VARCHAR(254)	NOT NULL				
phone	VARCHAR(16)	NOT NULL	' '			
subject	VARCHAR(200)	NOT NULL				
message	TEXT	NOT NULL				
category	VARCHAR(32)	NOT NULL	'general'			general order product partnership press
status	VARCHAR(16)	NOT NULL	'new'			new read responded closed spam
responded_by_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
responded_at	TIMESTAMPTZ	NULL				
ip_address	INET	NULL				Spam triage
created_at	TIMESTAMPTZ	NOT NULL	now()			UTC
updated_at	TIMESTAMPTZ	NOT NULL	now()			UTC; touched on write

15 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Contact-form submissions. The frontend has a Contact page with nowhere to post to.
Relationships	Optional reference to a signed-in user; may be escalated into a support_ticket.
Used By APIs	POST /contact (new)
Used By Frontend Pages	Contact Us page, Admin → Messages

support_ticket [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
ticket_number	VARCHAR(24)	NOT NULL		UQ		Customer-facing reference
user_id	BIGINT	NOT NULL		FK	users_user	ON DELETE CASCADE
order_id	BIGINT	NULL		FK	orders_order	ON DELETE SET NULL
return_request_id	BIGINT	NULL		FK	return_request	ON DELETE SET NULL
category	VARCHAR(32)	NOT NULL	'general'			order payment return product account general
subject	VARCHAR(200)	NOT NULL				
priority	VARCHAR(12)	NOT NULL	'normal'			low normal high urgent
status	VARCHAR(20)	NOT NULL	'open'			open in_progress waiting_customer resolved closed
assigned_to_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL

Column Name	Data Type	Nullable	Default	Key	References	Notes
first_response_at	TIMESTAMP TZ	NULL				SLA numerator
resolved_at	TIMESTAMP TZ	NULL				
sla_due_at	TIMESTAMP TZ	NULL				Breach reporting
satisfaction_rating	SMALLINT	NULL				CHECK 1..5
created_at	TIMESTAMP TZ	NOT NULL	now()			UTC
updated_at	TIMESTAMP TZ	NOT NULL	now()			UTC; touched on write

17 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Customer support cases with SLA tracking.
Relationships	Child of User; optionally linked to an order or return. Parent of support_message.
Used By APIs	GET POST /support/tickets · /support/tickets/{number} (new)
Used By Frontend Pages	Help Centre → My Tickets, Admin → Support Queue

support_message [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
uuid	UUID	NOT NULL	gen_random_uuid()	UQ		Public-safe identifier
ticket_id	BIGINT	NOT NULL		FK	support_ticket	ON DELETE CASCADE
sender_id	BIGINT	NULL		FK	users_user	ON DELETE SET NULL
sender_type	VARCHAR(12)	NOT NULL	'customer'			customer agent system
message	TEXT	NOT NULL				
attachment_url	VARCHAR(500)	NOT NULL	' '			
is_internal	BOOLEAN	NOT NULL	FALSE			Staff-only note
is_read	BOOLEAN	NOT NULL	FALSE			
created_at	TIMESTAMP TZ	NOT NULL	now()			UTC
updated_at	TIMESTAMP TZ	NOT NULL	now()			UTC; touched on write

11 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	One message in a ticket thread. is_internal keeps staff notes out of the customer view.
Relationships	Child of support_ticket (CASCADE).
Used By APIs	Embedded in ticket endpoints
Used By Frontend Pages	Ticket thread, Admin → Ticket Detail

ANALYTICS AGGREGATION

analytics_daily_sales [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
snapshot_date	DATE	NOT NULL		UQ*		Composite UQ with currency
currency	CHAR(3)	NOT NULL	'INR'	UQ*		
orders_count	INTEGER	NOT NULL	0			
units_sold	INTEGER	NOT NULL	0			
gross_revenue	NUMERIC(14,2)	NOT NULL	0.00			
discount_total	NUMERIC(14,2)	NOT NULL	0.00			
coupon_discount_total	NUMERIC(14,2)	NOT NULL	0.00			
shipping_total	NUMERIC(14,2)	NOT NULL	0.00			
tax_total	NUMERIC(14,2)	NOT NULL	0.00			
refund_total	NUMERIC(14,2)	NOT NULL	0.00			
net_revenue	NUMERIC(14,2)	NOT NULL	0.00			gross – refunds
average_order_value	NUMERIC(12,2)	NOT NULL	0.00			
cancelled_orders	INTEGER	NOT NULL	0			
new_customers	INTEGER	NOT NULL	0			
returning_customers	INTEGER	NOT NULL	0			
computed_at	TIMESTAMPTZ	NOT NULL	now()			Recompute stamp

17 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Pre-aggregated daily sales. The analytics module computes everything live from the orders table, which is correct now and will not hold past roughly a hundred thousand orders. One row per day per currency.
Relationships	Derived from orders_order and payments_refund by a nightly job. Rebuildable at any time from source — never the system of record.
Used By APIs	GET /admin/dashboard/summary · /admin/dashboard/charts · /admin/reports/revenue
Used By Frontend Pages	Admin Dashboard cards and charts, Revenue Report

analytics_product_daily [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
snapshot_date	DATE	NOT NULL		UQ*		Composite UQ with product_id
product_id	BIGINT	NOT NULL		FK/UQ*	products_product	ON DELETE CASCADE
views	INTEGER	NOT NULL	0			
unique_viewers	INTEGER	NOT NULL	0			
cart_adds	INTEGER	NOT NULL	0			
wishlist_adds	INTEGER	NOT NULL	0			
units_sold	INTEGER	NOT NULL	0			
revenue	NUMERIC(14,2)	NOT NULL	0.00			

Column Name	Data Type	Nullable	Default	Key	References	Notes
returns_count	INTEGER	NOT NULL	0			
conversion_rate	NUMERIC(5,2)	NOT NULL	0.00			purchases ÷ views
cart_abandonment_rate	NUMERIC(5,2)	NOT NULL	0.00			
computed_at	TIMESTAMPTZ	NOT NULL	now()			

13 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Per-product daily performance, and the funnel the current schema cannot produce: views to cart-adds to purchases.
Relationships	Child of products_product (CASCADE). Partition by snapshot_date range.
Used By APIs	GET /admin/analytics/top-products · /admin/reports/sales
Used By Frontend Pages	Admin Product Analytics, Sales Report

analytics_customer_daily [PROPOSED]

Column Name	Data Type	Nullable	Default	Key	References	Notes
id	BIGSERIAL	NOT NULL	identity	PK		
snapshot_date	DATE	NOT NULL		UQ		
total_customers	INTEGER	NOT NULL	0			
new_signups	INTEGER	NOT NULL	0			
active_customers	INTEGER	NOT NULL	0			Placed an order that day
returning_customers	INTEGER	NOT NULL	0			
churned_customers	INTEGER	NOT NULL	0			No order in 180 days
average_lifetime_value	NUMERIC(12,2)	NOT NULL	0.00			
repeat_purchase_rate	NUMERIC(5,2)	NOT NULL	0.00			
cart_abandonment_rate	NUMERIC(5,2)	NOT NULL	0.00			
computed_at	TIMESTAMPTZ	NOT NULL	now()			

11 columns · NEW BUILD REQUIRED

Attribute	Detail
Purpose	Customer cohort and retention metrics, computed once rather than on every dashboard load.
Relationships	Derived from users_user and orders_order nightly.
Used By APIs	GET /admin/dashboard/customers · /admin/reports/customers
Used By Frontend Pages	Admin Customer Analytics, Customer Report

ENTITY RELATIONSHIP DIAGRAM

Customer & Transaction Domain

```

users_user (root of the customer domain)
|
+-- users_address                1:N  one is_default per user (partial UQ)
+-- users_user_groups -----> auth_group
+-- users_user_user_permissions --> auth_permission
+-- admin_user_role -----> admin_role --> admin_role_permission --> admin_permission
|
+-- wishlist_wishlist            1:1
|   \-- wishlist_wishlistitem    1:N  --> products_product
|
+-- cart_cart                    1:N  (one ACTIVE per user – partial UQ)
|   \-- cart_cartitem            1:N  --> products_product
|                                   --> products_productvariant
|
+-- orders_order                 1:N  ON DELETE PROTECT
|   +-- orders_orderitem         1:N  --> products_product (SET NULL)
|   |                               --> products_productvariant (SET NULL)
|   |   +-- reviews_review       1:1  UNIQUE(order_item)
|   |   \-- return_item          1:N
|   +-- orders_orderstatushistory 1:N  append-only
|   +-- orders_shipment          1:N  --> delivery_partner
|   |   \-- shipment_tracking_event 1:N  append-only
|   +-- payments_payment         1:N  ON DELETE PROTECT
|   |   +-- payments_paymentattempt 1:N
|   |   +-- payments_refund        1:N  ON DELETE PROTECT
|   |   \-- payments_paymentwebhooklog 1:N  UNIQUE(gateway, event_id)
|   +-- coupons_couponusage      1:N  --> coupons_coupon
|   +-- return_request           1:N  --> payments_refund
|   |   +-- return_item          1:N
|   |   \-- exchange_request     1:N  --> products_productvariant
|   +-- loyalty_transaction      1:N
|   \-- gift_card_transaction    1:N
|
+-- reviews_review              1:N
|   +-- reviews_reviewimage      1:N
|   \-- reviews_helpfulvote     1:N
+-- reviews_helpfulvote         1:N  UNIQUE(review, user)
+-- product_question --> product_answer 1:N
|
+-- recommendations_recentlyviewed 1:N  user XOR session (CHECK)
+-- search_history               1:N
+-- clickstream_event            1:N  PARTITIONED BY RANGE(occurred_at)
+-- recommendation_history       1:N
+-- fashion_preference           1:1
|
+-- notifications_notification    1:N
+-- notifications_notificationpreference 1:1
+-- activity_log                 1:N  PARTITIONED BY RANGE(created_at)
|
+-- loyalty_account              1:1
|   \-- loyalty_transaction      1:N
+-- gift_card                    (purchased_by / redeemed_by)
+-- referral                     (referrer / referred_user – self-referencing)
|
\-- support_ticket               1:N
    \-- support_message          1:N

```

ENTITY RELATIONSHIP DIAGRAM

Catalog & Product Domain

catalog_category (root of the merchandising domain)

- +-- catalog_subcategory 1:N UNIQUE(category, name)
- | \-- products_product 1:N ON DELETE PROTECT
- |
- +-- catalog_brand_categories --> catalog_brand M:N
- +-- catalog_collection_categories --> catalog_collection M:N
- +-- coupons_coupon_categories --> coupons_coupon M:N
- +-- homepage_banner 1:N
- \-- products_product 1:N ON DELETE PROTECT

products_product (widest table: 57 columns)

- +-- products_productimage 1:N partial UQ: one is_primary
- +-- products_productvariant 1:N UNIQUE(product, colour, size)
- | +-- variant_image 1:N
- | +-- inventory_stock 1:N --> warehouse
- | +-- inventory_transaction 1:N append-only ledger
- | +-- stock_alert 1:N
- | +-- cart_cartitem 1:N
- | \-- orders_orderitem 1:N SET NULL
- +-- products_productattribute 1:N UNIQUE(product, key)
- +-- products_productspecification 1:N UNIQUE(product, label)
- +-- products_product_tags --> products_producttag M:N
- +-- product_video 1:N
- +-- product_price_history 1:N append-only
- +-- flash_sale_item --> flash_sale M:N
- +-- lookbook_item --> lookbook M:N
- +-- coupons_coupon_products --> coupons_coupon M:N
- +-- wishlist_wishlistitem 1:N
- +-- reviews_review 1:N
- +-- recommendations_recentlyviewed 1:N
- +-- recommendations_productaffinity 1:N self-join, directed, CHECK not self
- +-- analytics_product_daily 1:N
- \-- product_question 1:N

Lookup dimension (all referenced FROM products_product / _variant / users_address)

lkp_color · lkp_size · lkp_material · lkp_occasion · lkp_gender

lkp_age_group · lkp_season · lkp_pattern · lkp_fit · lkp_country

Referential Integrity Notes

#	Note
1	PROTECT edges (users_user → orders_order, orders_order → payments_payment, catalog_* → products_product) are deliberate: they make destructive deletes fail loudly instead of silently erasing financial history.
2	SET NULL edges (orders_orderitem → products_product, payments_paymentwebhooklog → payments_payment) exist where the child must outlive the parent. An order line is a historical fact and keeps its own snapshot.
3	CASCADE edges are used only where the child has no meaning without its parent — gallery images, cart lines, review photographs.
4	Two tables (cart_cart, recommendations_recentlyviewed) carry an exclusive-or CHECK over (user_id, session_key). Every guest-capable feature follows this shape.

STORED PROCEDURES

Business logic currently lives in the Django service layer. The procedures below are specified for operations where a per-step round trip is the actual cost, or where atomicity must hold under concurrency regardless of which application connects. They are specifications, not implementations — no procedure exists yet.

#	Procedure Name	Purpose	Parameters	Tables Used	Business Logic
1	sp_place_order	Convert an active cart into an order: snapshot lines, reserve stock, apply coupon, compute money, deactivate the cart.	p_user_id BIGINT, p_cart_id BIGINT, p_shipping_address_id BIGINT, p_payment_method VARCHAR, p_coupon_code VARCHAR	cart_cart, cart_cartitem, orders_order, orders_orderitem, orders_orderstatushistory, products_productvariant, inventory_stock, inventory_transaction, coupons_couponusage	SERIALIZABLE. Locks each variant FOR UPDATE, verifies quantity_available, increments quantity_reserved, writes snapshot lines, records the coupon redemption and emits the initial status-history row. Rolls back entirely on any insufficient stock.
2	sp_reserve_stock	Reserve units for a variant at checkout without oversell.	p_variant_id BIGINT, p_warehouse_id BIGINT, p_quantity INTEGER, p_reference_id BIGINT, OUT p_success BOOLEAN	inventory_stock, inventory_transaction	Single conditional UPDATE ... WHERE quantity_on_hand - quantity_reserved >= p_quantity. Zero rows updated means insufficient stock. Read-then-write would lose the race.
3	sp_commit_stock	Convert a reservation into a permanent decrement on payment capture.	p_order_id BIGINT	orders_orderitem, inventory_stock, inventory_transaction, products_product	Decrements quantity_on_hand and quantity_reserved together, writes a 'sale' ledger row per line and increments products_product.purchase_count.
4	sp_release_stock	Return reserved units on cancellation, expiry or payment failure.	p_order_id BIGINT, p_reason VARCHAR	orders_orderitem, inventory_stock, inventory_transaction	Decrements quantity_reserved only. Idempotent: a second call on an already-released order is a no-op, because an abandoned checkout can be swept twice.
5	sp_apply_coupon	Validate a coupon against a cart and return the discount.	p_cart_id BIGINT, p_user_id BIGINT, p_code VARCHAR, OUT p_discount NUMERIC, OUT p_reason VARCHAR	coupons_coupon, coupons_couponusage, cart_cart, cart_cartitem, coupons_coupon_categories/_brands/_products	Checks window, active flag, minimum cart value, global and per-user limits, first-order-only and category/brand/product scope. Returns the reason on failure so the UI can explain the refusal.
6	sp_process_refund	Record a refund and adjust the payment atomically.	p_payment_id BIGINT, p_amount NUMERIC, p_reason VARCHAR, p_processed_by BIGINT	payments_payment, payments_refund, orders_order, return_request	Locks the payment FOR UPDATE, verifies refunded_amount + p_amount <= amount, inserts the refund and moves the payment to refunded or partially_refunded.
7	sp_recompute_product_rating	Recalculate a product's cached rating from APPROVED reviews only.	p_product_id BIGINT	reviews_review, products_product	Approved-only is the whole point: averaging pending rows lets spam move the number shoppers see before a human has looked at it.
8	sp_rebuild_product_affinity	Rebuild the co-purchase graph from order history.	p_min_co_purchases INTEGER DEFAULT 2	orders_order, orders_orderitem, recommendations_productaffinity	Self-joins order lines within an order, counts directed pairs, computes confidence as co-purchases ÷ the left product's order count, and replaces the table in one transaction. Pairs below the threshold are dropped as coincidence.
9	sp_aggregate_daily_sales	Populate the daily sales snapshot.	p_snapshot_date DATE	orders_order, payments_refund, analytics_daily_sales	Idempotent upsert on (snapshot_date, currency), so a re-run after a late-settling refund corrects rather than duplicates.
10	sp_aggregate_product_daily	Populate per-product daily performance including the view-to-purchase funnel.	p_snapshot_date DATE	clickstream_event, orders_orderitem, analytics_product_daily	Reads one partition of clickstream_event, not the whole table.
11	sp_expire_coupons	Deactivate coupons past their validity window.	none	coupons_coupon	Single UPDATE. Expiry is already derivable at read time; this makes the admin list, the analytics counts and the storefront agree on one answer.
12	sp_expire_loyalty_points	Expire points past their expiry date and adjust the balance.	p_as_of DATE	loyalty_transaction, loyalty_account	Writes negative 'expire' ledger rows oldest-first and recomputes the balance from the ledger rather than trusting the cached figure.
13	sp_merge_guest_session	Fold a guest cart, wishlist and browsing trail into an account on sign-in.	p_user_id BIGINT, p_session_key VARCHAR	cart_cart, cart_cartitem, recommendations_recentlyviewed, search_history	Quantities are summed and clamped to the per-line limit. Trail rows the account already holds are dropped, since the unique constraint would reject the move.
14	sp_generate_order_number	Allocate a collision-free human-readable order number.	OUT p_order_number VARCHAR	orders_order, a dedicated sequence	Sequence-backed rather than timestamp-plus-random: two orders placed in the same millisecond currently collide on the unique index.

#	Procedure Name	Purpose	Parameters	Tables Used	Business Logic
15	sp_low_stock_scan	Raise stock alerts for variants at or below their reorder level.	none	inventory_stock, stock_alert, notifications_notification	Inserts only where an unresolved alert does not already exist, so a persistently low SKU produces one alert rather than one per run.
16	sp_process_return	Advance a return through QC and restock what is sellable.	p_return_request_id BIGINT, p_qc_passed BOOLEAN, p_processed_by BIGINT	return_request, return_item, inventory_stock, inventory_transaction, payments_refund	On QC pass, restocks only lines flagged is_restockable and triggers the refund. On fail, records the reason and leaves stock untouched.
17	sp_calculate_customer_lifetime	Recompute lifetime value and tier for one customer.	p_user_id BIGINT	orders_order, payments_refund, loyalty_account	Net of refunds. Gross LTV overstates the value of a serial returner, which is exactly the customer a tier should not promote.
18	sp_purge_expired_sessions	Retention sweep across guest-scoped and behavioural tables.	p_retention_days INTEGER DEFAULT 90	cart_cart, recommendations_recentlyviewed, search_history, clickstream_event	Drops whole partitions where the table is partitioned rather than issuing DELETES — a partition drop is a metadata operation and produces no bloat.
19	sp_audit_log_write	Generic trigger function capturing before/after images.	TG_OP, TG_TABLE_NAME, OLD, NEW (trigger context)	audit_log	Attached as an AFTER trigger on sensitive tables. Writes only the columns that actually changed, so an unrelated touch does not produce a diff nobody can read.
20	sp_dashboard_summary	Return every admin dashboard figure in one round trip.	p_days INTEGER DEFAULT 30	analytics_daily_sales, analytics_customer_daily, analytics_product_daily (all read-only)	Reads the aggregate tables, never the transactional ones. Replaces roughly fifteen separate aggregate queries.

DATABASE INDEX RECOMMENDATIONS

Table / Scope	Recommended Index	Rationale
Existing coverage		The shipped schema already carries 47 named composite indexes. They are documented per table earlier in this document and should not be duplicated.
products_product	GIN (to_tsvector('english', name ' ' short_description))	Product search currently uses chained ILIKE, which cannot use a B-tree. A GIN full-text index is the single largest available query win.
products_product	GIN (name gin_trgm_ops) – pg_trgm	Powers typo-tolerant autocomplete on /products/suggestions.
products_product	BRIN (created_at)	Append-mostly and correlated with physical order; a BRIN index is a fraction of the size of the B-tree for the same range scans.
orders_order	(user_id, status, created_at DESC)	Serves 'my orders filtered by status' without a sort.
orders_order	Partial: (created_at DESC) WHERE status = 'pending'	The abandoned-order sweep reads a tiny slice of a large table.
orders_orderitem	(product_id, order_id)	Analytics joins from product to order lines; currently only (order_id) is indexed.
cart_cart	Partial: (updated_at) WHERE is_active	Abandoned-cart job. Inactive carts dominate the table over time.
inventory_stock	(warehouse_id, variant_id) INCLUDE (quantity_available)	Covering index: allocation resolves without a heap fetch.
inventory_stock	Partial: (variant_id) WHERE quantity_available <= reorder_level	Low-stock scan touches only the rows that qualify.
clickstream_event	(user_id, occurred_at DESC) per partition	Local indexes on partitions; a global index would defeat partition pruning.
clickstream_event	GIN (metadata jsonb_path_ops)	Ad-hoc event property queries.
search_history	GIN (normalised_query gin_trgm_ops)	Fuzzy grouping of query variants for the trending job.
audit_log	(entity_type, entity_id, created_at DESC)	The only access pattern: the history of one record.
notifications_notification	Partial: (user_id, created_at DESC) WHERE NOT is_read	The unread badge is the hottest read in the module and touches a small subset.
reviews_review	Partial: (created_at) WHERE status = 'pending'	Moderation queue, oldest first.
recommendations_productaffinity	(product_id, score DESC) INCLUDE (related_product_id)	Covering index for the frequently-bought-together rail.
Extensions required	pg_trgm, btree_gin, pg_stat_statements, pgcrypto	pgcrypto supplies gen_random_uuid(); pg_stat_statements is how the slow-query list is produced at all.

PARTITION STRATEGY

PostgreSQL declarative partitioning. Retention becomes DROP PARTITION — a metadata operation that leaves no dead tuples and needs no VACUUM — which is the main reason to partition these tables at all.

Table	Partition Method	Interval	Retention	Rationale
clickstream_event	RANGE (occurred_at)	Daily	90 days hot, then to cold storage	Highest-volume table by an order of magnitude. Daily partitions keep each index small enough to stay in cache, and retention becomes DROP PARTITION — a metadata operation that produces no bloat and needs no VACUUM.
audit_log	RANGE (created_at)	Monthly	24 months (compliance)	Write-heavy, read-rarely, and legally required to be retained. Monthly partitions make the retention boundary a single DROP.
activity_log	RANGE (created_at)	Monthly	12 months	Same profile as audit_log at lower volume.
notifications_notification	RANGE (created_at)	Monthly	6 months	The retention job currently issues a bulk DELETE, which leaves dead tuples behind on a table the queue worker reads every two minutes.
inventory_transaction	RANGE (created_at)	Monthly	Indefinite (financial)	Append-only ledger. Old partitions are read only during audits and can be moved to slower storage.
search_history	RANGE (created_at)	Monthly	12 months	High volume, aggregated nightly and rarely read raw afterwards.
recommendation_history	RANGE (served_at)	Monthly	6 months	Written on every rail impression; useful only while a model version is live.
orders_order	RANGE (created_at) – deferred	Yearly	Indefinite	Do NOT partition at launch. Partitioning a table this heavily foreign-keyed forces every child to carry the partition key. Revisit past roughly five million orders.
analytics_product_daily	RANGE (snapshot_date)	Monthly	36 months	Grows at products × days. Partitioning keeps dashboard range scans bounded.

Operational note: create partitions ahead of time with pg_partman or a scheduled job. A missing partition rejects the insert, which on clickstream_event would drop instrumentation silently and on inventory_transaction would fail a checkout.

PERFORMANCE OPTIMIZATION SUGGESTIONS

Area	Current State	Recommendation
Full-text search	Product search chains ILIKE across name, description, brand and category. This cannot use an index and degrades linearly with catalogue size.	Add a generated tsvector column with a GIN index and rank with ts_rank_cd. Keep pg_trgm for typo tolerance on autocomplete.
Denormalised counters	Six counters on products_product (view_count, purchase_count, wishlist_count, rating_average, rating_count, review_count) are updated with F() expressions. Correct, but view_count writes contend on hot rows.	Batch view counts through Redis and flush periodically. The number is a merchandising signal, not an invoice — a few seconds of staleness costs nothing.
Analytics on live tables	Every dashboard figure aggregates orders_order and orders_orderitem directly. Fine at current volume; it will not hold past ~100k orders.	Introduce the three analytics_* snapshot tables and point the dashboard at them. The warm_analytics job already exists and becomes the writer.
N+1 prevention	The application already uses select_related and prefetch_related consistently (with_card_data, with_detail, with_recipient), and vote state is annotated with one Exists subquery per page.	Preserve this on new endpoints. Add a CI assertion on query counts for the listing and detail endpoints so a regression fails the build rather than production.
Connection pooling	CONN_MAX_AGE is 60s with health checks. Each Unicorn worker holds its own connection; workers × replicas can exhaust max_connections.	Put PgBouncer in transaction mode in front of the database. Note that transaction pooling forbids session-level features — the application uses none.
Covering indexes	Hot lookups fetch the heap for one or two extra columns.	Use INCLUDE on the frequently-bought-together and inventory allocation indexes to make them index-only scans.
JSONB columns	orders_order.shipping_address, payments_payment.raw_response and notifications_notification.context are JSONB.	Snapshots are correct as JSONB and should not be normalised. Add GIN indexes only where a JSONB column is actually filtered — today none are.
Bulk operations	The affinity rebuild already uses bulk_create with a batch size and streams source rows with an iterator.	Apply the same pattern to the analytics aggregation jobs. Use COPY for any import above roughly ten thousand rows.
Read replicas	Analytics and reports compete with checkout for the same database.	Route /admin/analytics/* and /admin/reports/* to a read replica. They tolerate replica lag; checkout does not.
Partial indexes	Several hot queries touch a small subset of a large table — unread notifications, pending reviews, active carts, low stock.	Partial indexes on those predicates are a fraction of the size and stay cached.
VACUUM tuning	High-churn tables (cart_cartitem, notifications_notification, inventory_stock) accumulate dead tuples faster than the default autovacuum threshold reacts.	Lower autovacuum_vacuum_scale_factor to 0.02 on those tables specifically.

NORMALIZATION REPORT

Normal Form	Verdict	Analysis
1NF — Atomic values, no repeating groups	SATISFIED	Every column holds a single value. Repeating groups are child tables: product images, variants, attributes, specifications, order lines and review photographs each have their own table rather than numbered columns. Three JSONB columns are deliberate exceptions and are not 1NF violations: orders_order.shipping_address and billing_address are immutable point-in-time snapshots, never queried by sub-field; payments_payment.raw_response and payments_paymentwebhooklog.payload are opaque third-party documents retained for dispute evidence; notifications_notification.context is a rendered audit copy. Normalising any of them would create a second version of a fact that must never change.
2NF — No partial dependency on a composite key	SATISFIED	Every table uses a single-column surrogate BIGSERIAL primary key, so no non-key attribute can depend on part of a key. Composite uniqueness — (product, colour, size), (cart, variant), (coupon, order), (gateway, event_id), (review, user) — is expressed as UNIQUE constraints alongside the surrogate key rather than as the key itself. Bridge tables (products_product_tags, coupons_coupon_brands, admin_role_permission) carry only their two foreign keys and, where meaningful, a grant timestamp — no attribute that depends on one side alone.
3NF — No transitive dependency	SATISFIED WITH DOCUMENTED DENORMALISATION	The normalised core is in 3NF. Six categories of denormalisation are deliberate, and each is maintained by a named service function with a recount routine: 1. Engagement counters on products_product (rating_average, rating_count, review_count, wishlist_count, view_count, purchase_count) are derivable from child tables. Storing them turns every listing sort into an index scan instead of a six-way aggregate join. Rebuilt by sp_recompute_product_rating. 2. Money columns on cart_cartitem and orders_orderitem are computable from price and quantity. Stored so the total a customer agreed to is a fact, not a recomputation that could disagree after a price change. 3. Order line snapshots (product_name, brand_name, sku, size, colour, image_url) duplicate the product tables. This is not redundancy — it is temporal correctness. The product FK is SET NULL precisely so a delisted product cannot rewrite history. 4. reviews_review.helpful_count duplicates a COUNT over reviews_helpfulvote. Stored so the review list can order by helpfulness without a join; rebuilt by recount_helpful. 5. products_productvariant.stock becomes a roll-up of inventory_stock once warehousing lands. The per-warehouse rows are authoritative. 6. The analytics_* tables are wholly derived and rebuildable from source at any time. They are a cache with a schema, never a system of record. Enum-backed CharFields (gender, material, occasion, season, fit, pattern, size) are a 3NF weakness in the shipped schema: the label is functionally dependent on the code but no relation holds it. The ten proposed lkp_* tables resolve this.
BCNF — Every determinant is a candidate key	SATISFIED	No table has overlapping candidate keys with a non-trivial functional dependency between them. The composite unique constraints are all over foreign keys and discriminators, none of which determine another attribute in the same table.
4NF — No multi-valued dependency	SATISFIED	Independent many-to-many relationships are separate bridge tables. A coupon's category scope, brand scope and product scope live in three tables rather than one, so the combinations are not multiplied.

FINAL DATABASE STATISTICS

Metric	Value	Detail
Total Tables	105	Implemented plus proposed; excludes framework tables
Implemented Tables	41	Present in the shipped backend and verified by introspection
Proposed Tables	64	New build required by the module brief
Framework Tables	8	django_migrations, django_content_type, django_session, auth_group, auth_permission, auth_group_permissions, token_blacklist_outstandingtoken, token_blacklist_blacklistedtoken
Grand Total Objects	113	Everything the database will physically contain
Master Tables	24	users_user, products_product, catalog_*, coupons_coupon, warehouse, influencer, cms_page ...
Transaction Tables	22	orders_*, payments_*, cart_*, return_*, gift_card_transaction, support_ticket ...
Lookup / Reference Tables	16	lkp_* dimension tables plus role, permission and partner masters
Bridge (Junction) Tables	14	Pure many-to-many resolvers
Ledger / Append-Only Tables	13	Never updated; corrections are new rows
Aggregate / Snapshot Tables	3	analytics_daily_sales, analytics_product_daily, analytics_customer_daily
Total Columns	1265	507 implemented + 758 proposed
Widest Table	products_product (57 columns)	Catalogue master
Total Stored Procedures	20	Specified; none implemented yet
Total Views	6	Recommended: v_product_catalog, v_order_summary, v_customer_ltv, v_inventory_status, v_review_summary, v_daily_revenue
Existing Indexes	47	Named composite indexes already in the migration state
Recommended Indexes	18	Full-text, trigram, partial, covering and BRIN
Total Indexes (target)	65	Excluding primary keys and unique constraints
Implemented Foreign Keys	63	Including bridge tables
Proposed Foreign Keys	71	Including lookup normalisation
Check Constraints	24 existing	Money floors, quantity bounds, rating range, date-window ordering, exclusive-or ownership
Unique Constraints	38 existing	Including 11 partial (filtered) unique indexes
Partitioned Tables	9	See Partition Strategy
Required Extensions	4	pg_trgm, btree_gin, pgcrypto, pg_stat_statements
API Endpoints Served	172	Verified against the generated OpenAPI 3.1 schema
Backend Test Coverage	925 tests	All passing at time of writing

APPENDIX A — FRAMEWORK TABLES

Created and managed by Django and SimpleJWT. Listed so the database team can account for every object in the database, but they are not part of the application data model and must not be altered by hand.

Table	Purpose	Notes
django_migrations	Applied migration ledger	Never modify. Rewriting a row makes the schema and the code disagree silently.
django_content_type	Model registry for generic relations and permissions	Managed
django_session	Server-side session store	Moves to Redis when REDIS_URL is set; the table remains for the DB fallback.
auth_group	Django role groups	Superseded by admin_role for back-office RBAC
auth_permission	Per-model CRUD permissions	Auto-generated per model
auth_group_permissions	Bridge: group ↔ permission	Managed
token_blacklist_outstandingtoken	Issued JWT refresh tokens	Grows with every login; prune on a schedule.
token_blacklist_blacklistedtoken	Revoked refresh tokens	Consulted on every refresh — keep indexed and pruned.

APPENDIX B — LEGEND & COLOUR CODING

Row Colour	Meaning
Yellow	Primary key column
Green	Foreign key column
Blue	Unique-constrained column
Orange (Overview)	Proposed table — new build required
Alternating grey	Regular column

Key Marker	Meaning
PK	Primary key
FK	Foreign key
UQ	Unique constraint
PK/FK	Primary key that is also a foreign key
FK/UQ	Foreign key carrying a one-to-one uniqueness rule
UQ*	Part of a composite unique constraint

Data Type	Usage
BIGSERIAL	Surrogate primary key; identity
UUID	Public-safe secondary identifier, gen_random_uuid()
NUMERIC(12, 2)	Money. Never FLOAT.
NUMERIC(14, 2)	Aggregated money in analytics tables
TIMESTAMPTZ	All timestamps, stored UTC
JSONB	Immutable snapshots and opaque third-party payloads only
INET	IP addresses; truncate or hash under DPDP retention rules
CHAR(3)	ISO 4217 currency code
GENERATED ... STORED	Computed column materialised on disk